

MODULE 3

4.2 BUSINESS INTELLIGENCE AND DATA WAREHOUSING

Business intelligence (BI), as a term to describe evidence/fact-based managerial decision making, has been around for more than 20 years. With the emergence of business analytics as a new buzzword to describe pretty much the same managerial phenomenon, the popularity of BI as a term has gone down. As opposed to being an all-encompassing term, nowadays BI is used to describe the early stages of business analytics (i.e., descriptive analytics).

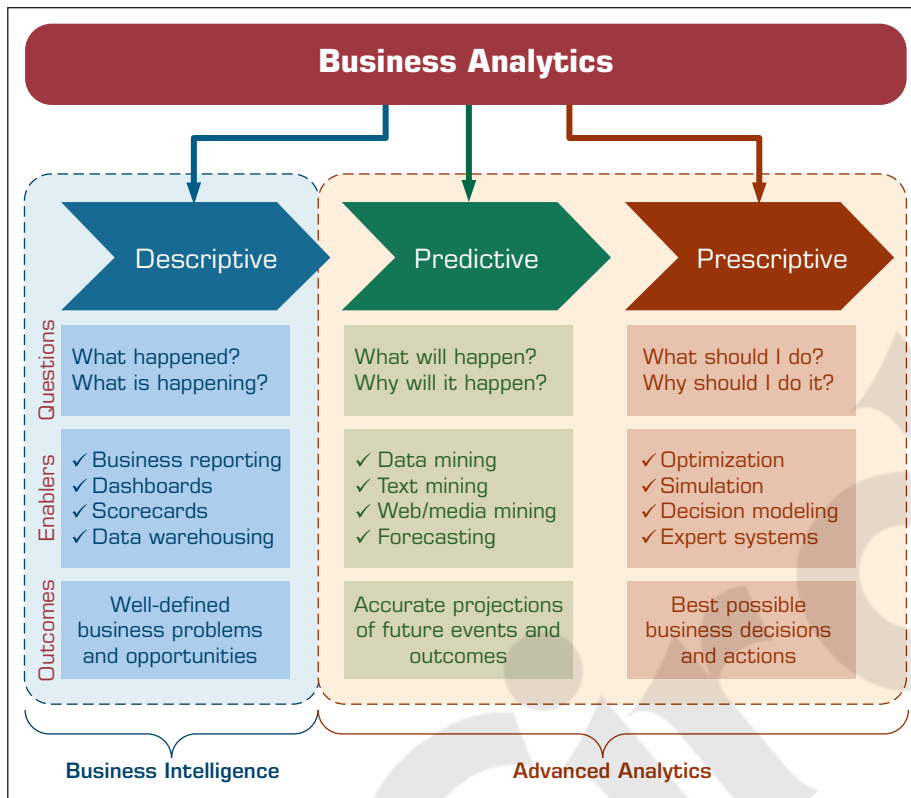


FIGURE 4.1 Relationship between Business Analytics and BI, and BI and Data Warehousing.

Figure 4.1 (a simplified version of which was shown and described in Chapter 1 to describe business analytics taxonomy) illustrates the relationship between BI and business analytics from a conceptual perspective. As shown therein, BI is the descriptive analytics portion of the business analytics continuum, the maturity of which leads to advanced analytics—a combination of predictive and prescriptive analytics.

Descriptive analytics (i.e., BI) is the entry level in the business analytics taxonomy. It is often called business reporting because of the fact that most of the analytics activities at this level deal with creating reports to summarize business activities to answer questions such as “What happened?” and “What is happening?” The spectrum of these reports includes static snapshots of business transactions delivered to knowledge workers (i.e., decision makers) on a fixed schedule (e.g., daily, weekly, quarterly); ad hoc reporting where the decision maker is given the capability of creating his or her own specific report (using an intuitive drag-and-drop graphical user interface) to address a specific or unique decision situation; and dynamic views of key business performance indicators (often captured and presented within a business performance management system) delivered to managers and executives in an easily digestible form (e.g., dashboard-looking graphical interfaces) on a continuous manner.

Generally speaking, and as depicted in Figure 4.1, BI systems rely on a data warehouse as the information source for creating insight and supporting managerial decisions. A multitude of organizational and external data is captured, transformed, and stored in a data warehouse to support timely and accurate decisions through enriched business insight. This chapter aims to cover the concepts, methods, and tools related to data warehousing and business performance management.

What Is a Data Warehouse?

In simple terms, a **data warehouse (DW)** is a pool of data produced to support decision making; it is also a repository of current and historical data of potential interest to managers throughout the organization. Data are usually structured to be available in a form ready for analytical processing activities (i.e., online analytical processing [OLAP], data mining, querying, reporting, and other decision support applications). A data warehouse is a subject-oriented, integrated, time-variant, nonvolatile collection of data in support of management's decision-making process.

A Historical Perspective to Data Warehousing

Even though *data warehousing* is a relatively new term in information technology (IT), its roots can be traced back in time, even before computers were widely used. In the early 1900s, people were using data (though mostly via manual methods) to formulate trends to help business users make informed decisions, which is the most prevailing purpose of data warehousing.

The motivations that led to the development of data warehousing technologies go back to the 1970s, when the computing world was dominated by mainframes. Real business data-processing applications, the ones run on the corporate mainframes, had complicated file structures using early-generation databases (not the table-oriented relational databases most applications use today) in which they stored data. Although these applications did a decent job of performing routine transactional data-processing functions, the data created as a result of these functions (such as information about customers, the products they ordered, and how much money they spent) was locked away in the depths of the files and databases. When aggregated information such as sales trends by region and by product type was needed, one had to formally request it from the data-processing department, where it was put on a waiting list with a couple of hundred other report requests (Hammergren & Simon, 2009). Even though the need for information and the data used to generate it existed, the database technology was not there to satisfy it. Figure 4.2 shows a timeline where some of the significant events that led to the development of data warehousing are shown.

Later in the last century, commercial hardware and software companies began to emerge with solutions to this problem. Between 1976 and 1979, the concept for a new company, Teradata, grew out of research at the California Institute of Technology

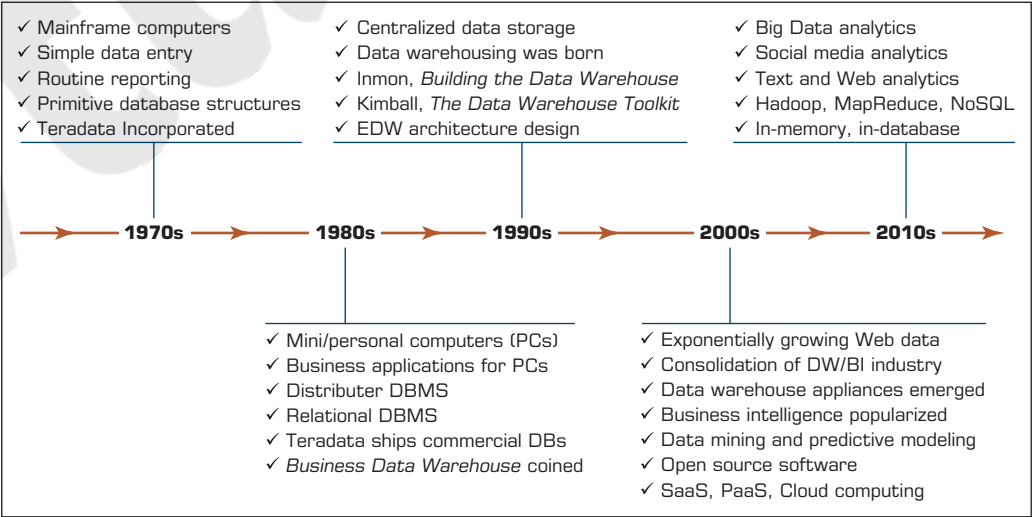


FIGURE 4.2 A List of Events That Led to Data Warehousing Development.

(Caltech), driven from discussions with Citibank's advanced technology group. Founders worked to design a database management system for parallel processing with multiple microprocessors, targeted specifically for decision support. Teradata was incorporated on July 13, 1979, and started in a garage in Brentwood, California. The name *Teradata* was chosen to symbolize the ability to manage terabytes (trillions of bytes) of data.

The 1980s were the decade of personal computers and minicomputers. Before anyone knew it, real computer applications were no longer only on mainframes; they were all over the place—everywhere you looked in an organization. That led to a portentous problem called *islands of data*. The solution to this problem led to a new type of software, called a *distributed database management system*, which would magically pull the requested data from databases across the organization, bring all the data back to the same place, and then consolidate it, sort it, and do whatever else was necessary to answer the user's question. Although the concept was a good one and early results from research were promising, the results were plain and simple: They just didn't work efficiently in the real world, and the islands-of-data problem still existed.

Meanwhile, Teradata began shipping commercial products to solve this problem. Wells Fargo Bank received the first Teradata test system in 1983, a parallel RDBMS (relational database management system) for decision support—the world's first. By 1984, Teradata released a production version of their product, and in 1986, *Fortune* magazine named Teradata "Product of the Year." Teradata, still in existence today, built the first data warehousing appliance—a combination of hardware and software to solve the data warehousing needs of many. Other companies began to formulate their strategies, as well.

During the 1980s, several other events occurred, collectively making it the decade of data warehousing innovation. For instance, Ralph Kimball founded Red Brick Systems in 1986. Red Brick began to emerge as a visionary software company by discussing how to improve data access; in 1988, Barry Devlin and Paul Murphy of IBM Ireland introduced the term *business data warehouse* as a key component of business information systems.

In the 1990s, a new approach to solving the islands-of-data problem surfaced. If the 1980s approach of reaching out and accessing data directly from the files and databases didn't work, the 1990s philosophy involved going back to the 1970s, in which data from those places was copied to another location—only doing it right this time; hence, data warehousing was born. In 1993, Bill Inmon wrote the seminal book *Building the Data Warehouse*. Many people recognize Inmon as the father of data warehousing. Additional publications emerged, including the 1996 book by Ralph Kimball, *The Data Warehouse Toolkit*, which discussed general-purpose dimensional design techniques to improve the data architecture for query-centered decision support systems.

In the 2000s, in the world of data warehousing, both popularity and the amount of data continued to grow. The vendor community and options began to consolidate. In 2006, Microsoft acquired ProClarity, jumping into the data warehousing market. In 2007, Oracle purchased Hyperion, SAP acquired Business Objects, and IBM merged with Cognos. The data warehousing leaders of the 1990s have been swallowed by some of the largest providers of information system solutions in the world. During this time, other innovations emerged, including data warehouse appliances from vendors such as Netezza (acquired by IBM), Greenplum (acquired by EMC), DATAlegro (acquired by Microsoft), and performance management appliances that enabled real-time performance monitoring. These innovative solutions provided cost savings because they were plug-compatible to legacy data warehouse solutions.

Since 2010, the big buzz has been *Big Data*. Many believe that Big Data is going to make an impact on data warehousing as we know it. Either they will find a way to coexist (which seems to be the most likely case, at least for several years) or Big Data (and the technologies that come with it) will make traditional data warehousing obsolete. The technologies that came with Big Data include Hadoop, MapReduce, NoSQL, and Hive.

Maybe we will see a new term coined in the world of data that combines the needs and capabilities of traditional data warehousing and the Big Data phenomenon.

Analytics in Action 4.1 is all about enhancing customer experience (CX) in financial sector through use of all kinds of data obtained from all channels along with business analytics.

ANALYTICS IN ACTION 4.1 Data-Driven Customer Experience in Financial Services

Introduction

The rapid migration to digital banking technologies driven by the pandemic will continue into the recovery. One-third of retail banking customers plan to increase their use of online and mobile banking services post COVID-19. Financial Services enterprises that make the right investments now could build an enduring advantage in serving customers.

Advanced customer analytics informed by new digital banking sources allows financial services firms to develop insights on customer cohorts and behaviors, and to more precisely target and tailor products and value propositions. Recent advances in analytics, such as non-linear machine-learning algorithms, combined with new and more granular data and are dramatically improving model efficacy and resulting service improvements, revenue increases, and lowering the cost to serve. Activating insights through triggers derived from real-time behaviors further contribute to these business outcomes.

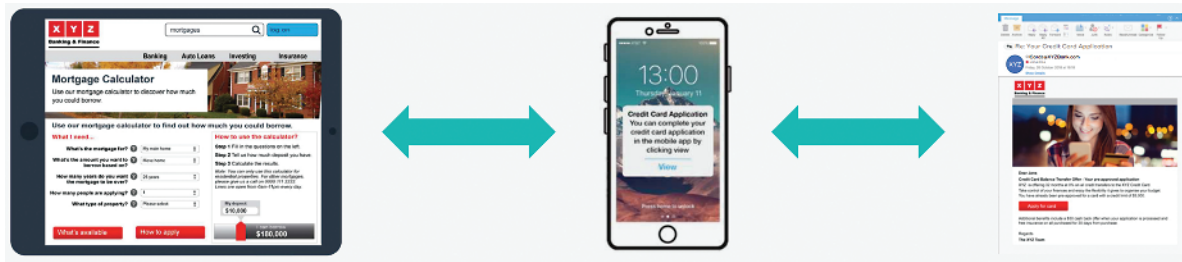
Challenges

Optimizing the customer experience can be challenging. For one thing, most organizations have little visibility into the end-to-end customer journey that spans multiple customer touchpoints across multiple departments. Furthermore, enterprises suffer from an inability to sense and react to a multitude of customer pain points and opportunities. Finally, organizations struggle to meaningfully personalize customer interactions in a timely manner, particularly in real-time, at the moments of truth.

Solutions

Vantage Customer Experience (Vantage CX) uniquely overcomes these challenges, enabling financial services providers to execute a sustainable customer experience initiative, including the following:

- Enables the previously elusive 360° view of the customer by providing financial services industry data models that include 100% of online behaviors and transactions. Granular, individual level behavioral data is captured in real time to provide an enhance view of every customer interaction across all digital properties. Online behavioral data is unified with customer transactions and behaviors from branch operations, call center, all transactions covering deposits, trades, transfers, applications, and more. This unified view of the customer across multiple departments and systems provides visibility into the full customer experience.
- Turns data into customer insights by putting the marketer and CX professional in control with powerful, easy-to use machine learning analytics along with other advanced analytics that provide richer, faster insights into customer behavior. No coding is required to leverage intuitive advanced analytics in Vantage, including Path for understanding the customer journey, Text for sentiment analysis, Cluster for hyper-segmentation, and Machine Learning to optimize the next best offer.
- Turns customer insights into action by delivering personalized customer interactions in real time, seamlessly across channels. Vantage CX enables real-time dynamic personalization by determining the optimal message to deliver based on both what the customer is doing right now combined with historical, integrated information on that customer's experiences and behaviors. Machine learning optimizes the goal selected by the business user,



such as maximizing revenue or minimizing cost to serve. These highly personalized offers are delivered at scale to reach millions of customers informed by billions of transactions and interactions. Real-time personalization makes it easy to deliver these dynamic messages through a company's Web site, SMS, POS, and other digital screens.

Highly detailed and structured Celebris data captured from all digital channels is fed into Vantage CX to enable highly effective customer journey analytics and real-time personalization at scale. Previously undiscovered issues with customer facing channels and pages can be quickly identified and resolved. Millisecond omnichannel data feeds combined with historical integrated data and machine learning insights from Vantage CX enable the creation of definitive and highly accurate customer profiles and the ability to orchestrate personalized customer experiences in-the-moment and across all channels.

Results

Working with Teradata and Celebris, the bank:

- Utilized identity resolution to connect web, mobile app, and offline customer data.
- Derived new digital variables such as page scores, recurring visits, duration, and others.
- Uncovered optimization opportunities across multiple touch points using Path analytics.
- Triggered hyper-personalized communications to recover abandoned applications.

In addition, the data and resultant insight enabled them to not only develop new products specifically designed for their digitally engaged customers based on enhanced customer segmentation, but also maximize acquisition rates and reduce service costs for those new offerings. Customer acquisition is now 3–5 times more effective overall than before for the same level of investment. Credit card conversion significantly improved for both the online and mobile channels: These successes fundamentally shifted the bank's digital banking department from product-centric to customer centric and accelerated the speed of the transformation program across the entire global organization.

Source: Teradata case study. Data-Driven Customer Experience in Financial Services. <https://www.teradata.com/Resources/Case-Studies/Data-Driven-Customer-Experience-in-Financial-Services> (accessed June 2022).

Characteristics of Data Warehousing

A common way to introduce data warehousing is to refer to its fundamental characteristics (see Inmon, 2005):

- **Subject oriented.** Data are organized by detailed subject, such as sales, products, or customers, containing only information relevant for decision support. Subject orientation enables users to determine not only how their business is performing, but also why. A data warehouse differs from an operational database in that most operational databases have a product orientation and are tuned to handle transactions that update the database. Subject orientation provides a more comprehensive view of the organization.

- **Integrated.** Integration is closely related to subject orientation. Data warehouses must place data from different sources into a consistent format. To do so, they must deal with naming conflicts and discrepancies among units of measure. A data warehouse is presumed to be totally integrated.
- **Time variant (time series).** A warehouse maintains historical data. The data do not necessarily provide current status (except in real-time systems). They detect trends, deviations, and long-term relationships for forecasting and comparisons, leading to decision making. Every data warehouse has a temporal quality. Time is the one important dimension that all data warehouses must support. Data for analysis from multiple sources contain multiple time points (e.g., daily, weekly, monthly views).
- **Nonvolatile.** After data are entered into a data warehouse, users cannot change or update the data. Obsolete data are discarded, and changes are recorded as new data.

These characteristics enable data warehouses to be tuned almost exclusively for data access. Some additional characteristics may include the following:

- **Web based.** Data warehouses are typically designed to provide an efficient computing environment for Web-based applications.
- **Relational/multidimensional.** A data warehouse uses either a relational structure or a multidimensional structure. A survey on multidimensional structures can be found in Romero and Abelló (2009).
- **Client/server.** A data warehouse uses the client/server architecture to provide easy access for end users.
- **Real time.** Newer data warehouses provide real-time, or active, data-access and analysis capabilities (see Basu, 2003; Bonde & Kuckuk, 2004).
- **Include metadata.** A data warehouse contains metadata (data about data) about how the data are organized and how to effectively use them.

Whereas a data warehouse is a repository of data, data warehousing is literally the entire process (see Watson, 2002). Data warehousing is a discipline that results in applications that provide decision support capability, allows ready access to business information, and creates business insight. The three main types of data warehouses are data marts (DMs), operational data stores (ODS), and enterprise data warehouses (EDW). In addition to discussing these three types of warehouses next, we also discuss metadata.

Data Marts

Whereas a data warehouse combines databases across an entire enterprise, a **data mart (DM)** is usually smaller and focuses on a particular subject or department. A DM is a subset of a data warehouse, typically consisting of a single subject area (e.g., marketing, operations). A DM can be either dependent or independent. A **dependent data mart** is a subset that is created directly from the data warehouse. It has the advantages of using a consistent data model and providing quality data. Dependent DMs support the concept of a single enterprise-wide data model, but the data warehouse must be constructed first. A dependent DM ensures that the end user is viewing the same version of the data that is accessed by all other data warehouse users. The high cost of data warehouses limits their use to large companies. As an alternative, many firms use a lower-cost, scaled-down version of a data warehouse referred to as an *independent DM*. An **independent data mart** is a small warehouse designed for a strategic business unit or a department, but its source is not an EDW.

Operational Data Stores

An **operational data store (ODS)** provides a fairly recent form of customer information file. This type of database is often used as an interim staging area for a data warehouse. Unlike the static contents of a data warehouse, the contents of an ODS are updated throughout the course of business operations. An ODS is used for short-term decisions

involving mission-critical applications rather than for the medium- and long-term decisions associated with an EDW. An ODS is similar to short-term memory in that it stores only very recent information. In comparison, a data warehouse is like long-term memory because it stores permanent information. An ODS consolidates data from multiple source systems and provides a near-real-time, integrated view of volatile, current data. The exchange, transfer, and load (ETL) processes (discussed later in this chapter) for an ODS are identical to those for a data warehouse. Finally, **oper marts** (see Imhoff, 2001) are created when operational data needs to be analyzed multidimensionally. The data for an oper mart come from an ODS.

Enterprise Data Warehouses (EDW)

An **enterprise data warehouse (EDW)** is a large-scale data warehouse that is used across the enterprise for decision support. The large-scale nature of an EDW provides integration of data from many sources into a standard format for effective BI and decision support applications. EDWs are used to provide data for many types of decision support systems (DSS), including customer relationship management (CRM), supply chain management (SCM), business performance management (BPM), business activity monitoring, product life cycle management, revenue management, and sometimes even knowledge management systems.

Metadata

Metadata are data about data (e.g., see Sen, 2004; Zhao, 2005). Metadata describe the structure of and some meaning about data, thereby contributing to their effective or ineffective use. Mehra (2005) indicated that few organizations really understand metadata, and fewer understand how to design and implement a metadata strategy. Metadata are generally defined in terms of usage as technical or business metadata. Patterns are another way to view metadata. According to the pattern view, we can differentiate between syntactic metadata (i.e., data describing the syntax of data), structural metadata (i.e., data describing the structure of the data), and semantic metadata (i.e., data describing the meaning of the data in a specific domain).

SECTION 4.2 REVIEW QUESTIONS

1. What is a data warehouse?
2. How does a data warehouse differ from a transactional database?
3. What is an ODS?
4. Differentiate among a DM, an ODS, and an EDW.
5. What is metadata? Explain the importance of metadata.

4.3 DATA WAREHOUSING PROCESS

Organizations, private and public, continuously collect data, information, and knowledge at an increasingly accelerated rate and store them in computerized systems. Maintaining and using these data and information becomes extremely complex, especially as scalability issues arise. In addition, the number of users needing to access the information continues to increase as a result of improved reliability and availability of network access, especially the Internet. Working with multiple databases, either integrated in a data warehouse or not, has become an extremely difficult task requiring considerable expertise, but it can provide immense benefits far exceeding its cost. As an illustrative example, Figure 4.3 shows business benefits of the EDW built by Teradata for a major automobile manufacturer.

Many organizations need to create data warehouses—massive data stores of time series data for decision support. Data are imported from various external and internal

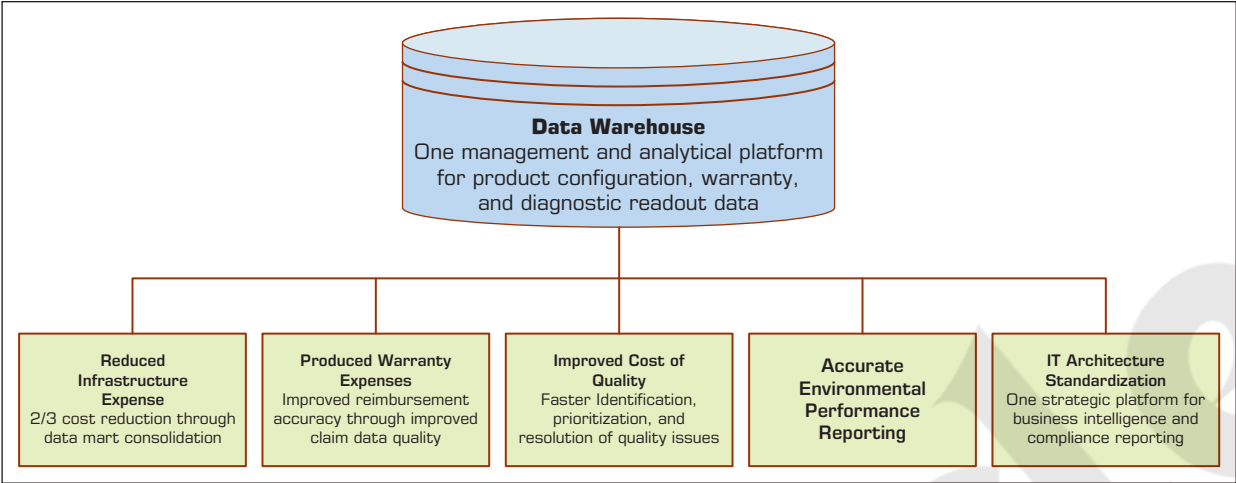


FIGURE 4.3 Data-Driven Decision Making—Business Benefits of the Data Warehouse. *Source: Teradata Corp.*

resources and are cleansed and organized in a manner consistent with the organization’s needs. After the data are populated in the data warehouse, DMs can be loaded for a specific area or department. Alternatively, DMs can be created first, as needed, and then integrated into an EDW. Often, though, DMs are not developed, but data are simply loaded onto PCs or left in their original state for direct manipulation using BI tools.

In Figure 4.4, we show a conceptual depiction of a typical data framework.

The following are the major components of the data warehousing process:

- **Data sources.** Data are sourced from multiple independent operational “legacy” systems and possibly from external data providers (such as the U.S. Census). Data may also come from an OLTP or enterprise resource planning (ERP) system. Web data in the form of Web logs may also feed to a data warehouse.
- **Data extraction and transformation.** Data are extracted and properly transformed using custom-written or commercial software called ETL.

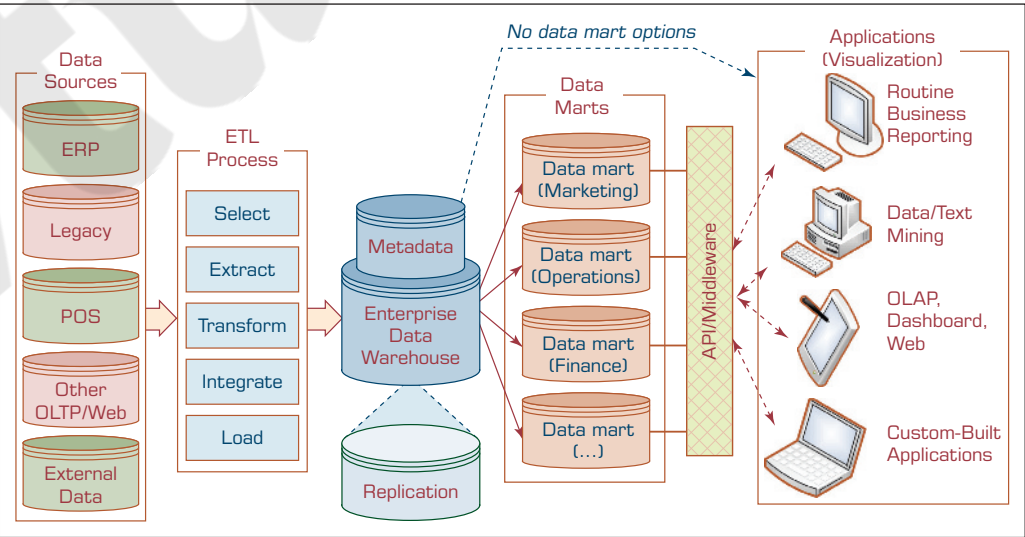


FIGURE 4.4 High-Level Depiction of a Data Warehouse Framework.

- **Data loading.** Data are loaded into a staging area, where they are transformed and cleansed. The data are then ready to load into the data warehouse and/or DMs.
- **Comprehensive database.** Essentially, this is the EDW to support all decision analysis by providing relevant summarized and detailed information originating from many different sources.
- **Metadata.** Metadata are maintained so that they can be assessed by IT personnel and users. Metadata include software programs about data and rules for organizing data summaries that are easy to index and search, especially with Web tools.
- **Middleware tools.** Middleware tools enable access to the data warehouse. Power users such as analysts may write their own SQL queries. Others may employ a managed query environment, such as Business Objects, to access data. There are many front-end applications that business users can use to interact with data stored in the data repositories, including data mining, OLAP, reporting tools, and data visualization tools.

SECTION 4.3 REVIEW QUESTIONS

1. Describe the data warehousing process.
2. Describe the major components of a data warehouse.
3. Identify and discuss the role of middleware tools.

4.4 DATA WAREHOUSING ARCHITECTURES

Several basic information system architectures can be used for data warehousing. Generally speaking, these architectures are commonly called client/server or n-tier architectures, of which two-tier and three-tier architectures are the most common (see Figures 4.5 and 4.6), but sometimes there is simply one tier. These types of multitiered architectures are known to be capable of serving the needs of large-scale, performance-demanding information systems such as data warehouses. Referring to the traditional use of n-tiered architectures for IT systems, we can distinguish among these architectures by dividing the data warehouse into three parts:

1. The data warehouse itself, which contains the data and associated software
2. Data acquisition (back-end) software, which extracts data from legacy systems and external sources, consolidates and summarizes them, and loads them into the data warehouse
3. Client (front-end) software, which allows users to access and analyze data from the warehouse (a DSS/BI/business analytics [BA] engine)

In a three-tier architecture, operational systems contain the data and the software for data acquisition in one tier (i.e., the server), the data warehouse is another tier, and the third tier includes the DSS/BI/BA engine (i.e., the application server) and the client (see Figure 4.5). Data from the warehouse are processed twice and deposited in an

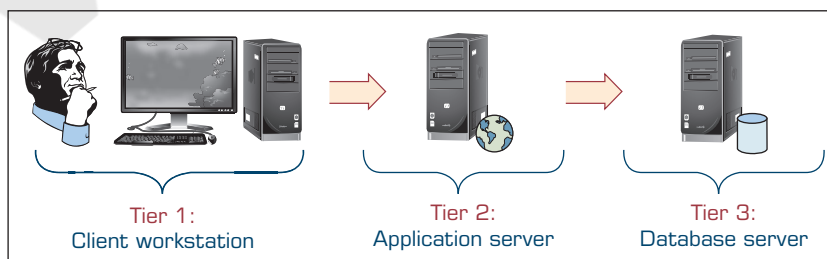


FIGURE 4.5 Architecture of a Three-Tier Data Warehouse.

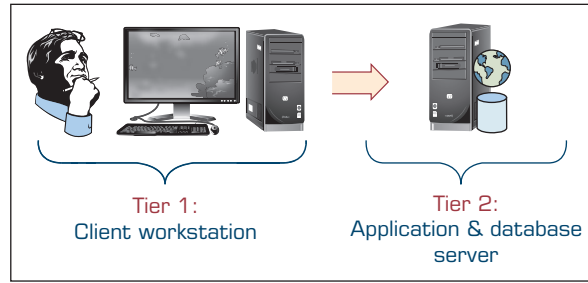


FIGURE 4.6 Architecture of a Two-Tier Data Warehouse.

additional multidimensional database, organized for easy multidimensional analysis and presentation, or replicated in DMs. The advantage of the three-tier architecture is its separation of the functions of the data warehouse, which eliminates resource constraints and makes it possible to easily create DMs.

In a two-tier architecture, the DSS engine physically runs on the same hardware platform as the data warehouse (see Figure 4.6). Therefore, it is more economical than the three-tier structure. The two-tier architecture can have performance problems for large data warehouses that work with data-intensive applications for decision support.

Much of the common wisdom assumes an absolutist approach, maintaining that one solution is better than the other, despite the organization's circumstances and unique needs. To further complicate these architectural decisions, many consultants and software vendors focus on one portion of the architecture, therefore limiting their capacity and motivation to assist an organization through the options based on its needs. But these aspects are being questioned and analyzed.

Data warehousing and the Internet are two key technologies that offer important solutions for managing corporate data. The integration of these two technologies produces Web-based data warehousing. In Figure 4.7, we show the architecture of Web-based data warehousing. The architecture is three-tiered and includes the PC client, Web server, and application server. On the client side, the user needs an Internet connection and a Web browser (preferably Java-enabled) through the familiar graphical user interface (GUI). The Internet/intranet/extranet is the communication medium between client

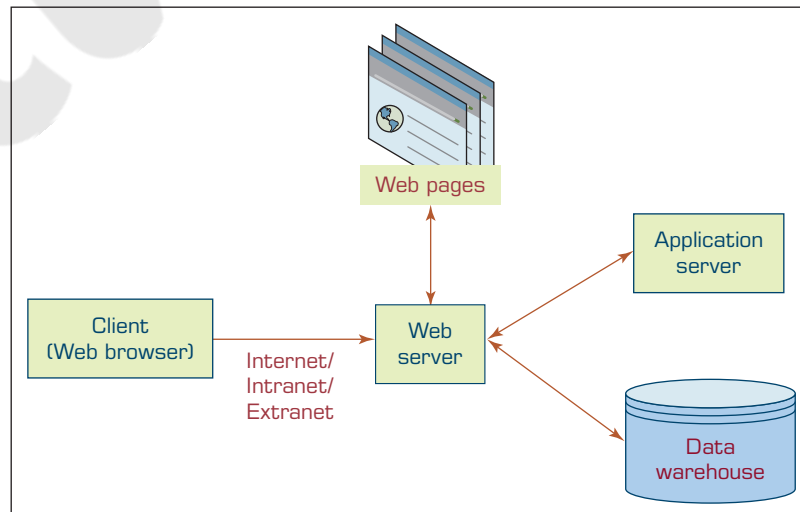


FIGURE 4.7 Architecture of Web-Based Data Warehousing.

and servers. On the server side, a Web server is used to manage the inflow and outflow of information between client and server. It is backed by both a data warehouse and an application server. Web-based data warehousing offers several compelling advantages, including ease of access, platform independence, and lower cost.

Web architectures for data warehousing are similar in structure to other data warehousing architectures, requiring a design choice for housing the Web data warehouse with the transaction server or as a separate server(s). Page-loading speed is an important consideration in designing Web-based applications; therefore, server capacity must be planned carefully.

Several issues must be considered when deciding which architecture to use. Among them are the following:

- ***Which database management system (DBMS) should be used?*** Most data warehouses are built using RDBMS. Oracle (Oracle Corporation, oracle.com), SQL Server (Microsoft Corporation, <http://microsoft.com/sql>), and DB2 (IBM Corporation, <http://www-01.ibm.com/software/data/db2>) are the ones most commonly used. Each of these products supports both client/server and Web-based architectures.
- ***Will parallel processing and/or partitioning be used?*** Parallel processing enables multiple central processing units (CPUs) to process data warehouse query requests simultaneously and provides scalability. Data warehouse designers need to decide whether the database tables will be partitioned (i.e., split into smaller tables) for access efficiency and what the criteria will be. This is an important consideration that is necessitated by the large amounts of data contained in a typical data warehouse. A recent survey on parallel and distributed data warehouses can be found in Furtado (2009). Teradata (teradata.com) has successfully adopted and is often commended on its novel implementation of this approach.
- ***Will data migration tools be used to load the data warehouse?*** Moving data from an existing system into a data warehouse is a tedious and laborious task. Depending on the diversity and the location of the data assets, migration may be a relatively simple procedure or (on the contrary) a months-long project. The results of a thorough assessment of the existing data assets should be used to determine whether to use migration tools, and if so, what capabilities to seek in those commercial tools.
- ***What tools will be used to support data retrieval and analysis?*** Often it is necessary to use specialized tools to periodically locate, access, analyze, extract, transform, and load necessary data into a data warehouse. A decision has to be made on (1) developing the migration tools in-house, (2) purchasing them from a third-party provider, or (3) using the ones provided with the data warehouse system. Overly complex, real-time migrations warrant specialized third-party ETL tools.

Alternative Data Warehousing Architectures

At the highest level, data warehouse architecture design viewpoints can be categorized into enterprise-wide data warehouse (EDW) design and DM design (Golfarelli & Rizzi, 2009). In Figure 4.8a–e, we show some alternatives to the basic architectural design types that are neither pure EDW nor pure DM, but in between or beyond the traditional architectural structures. Notable new ones include hub-and-spoke and federated architectures. The five architectures shown in Figure 4.8a–e, are proposed by Ariyachandra and Watson (2005, 2006a,b). Previously, in an extensive study, Sen and Sinha (2005) identified 15 different data warehousing methodologies. The sources of these methodologies are classified into three broad categories: core-technology vendors, infrastructure vendors, and information-modeling companies.

- ***Independent data marts.*** This is arguably the simplest and the least costly architecture alternative. The DMs are developed to operate independent of each other to serve the needs of individual organizational units. Because of their independence,

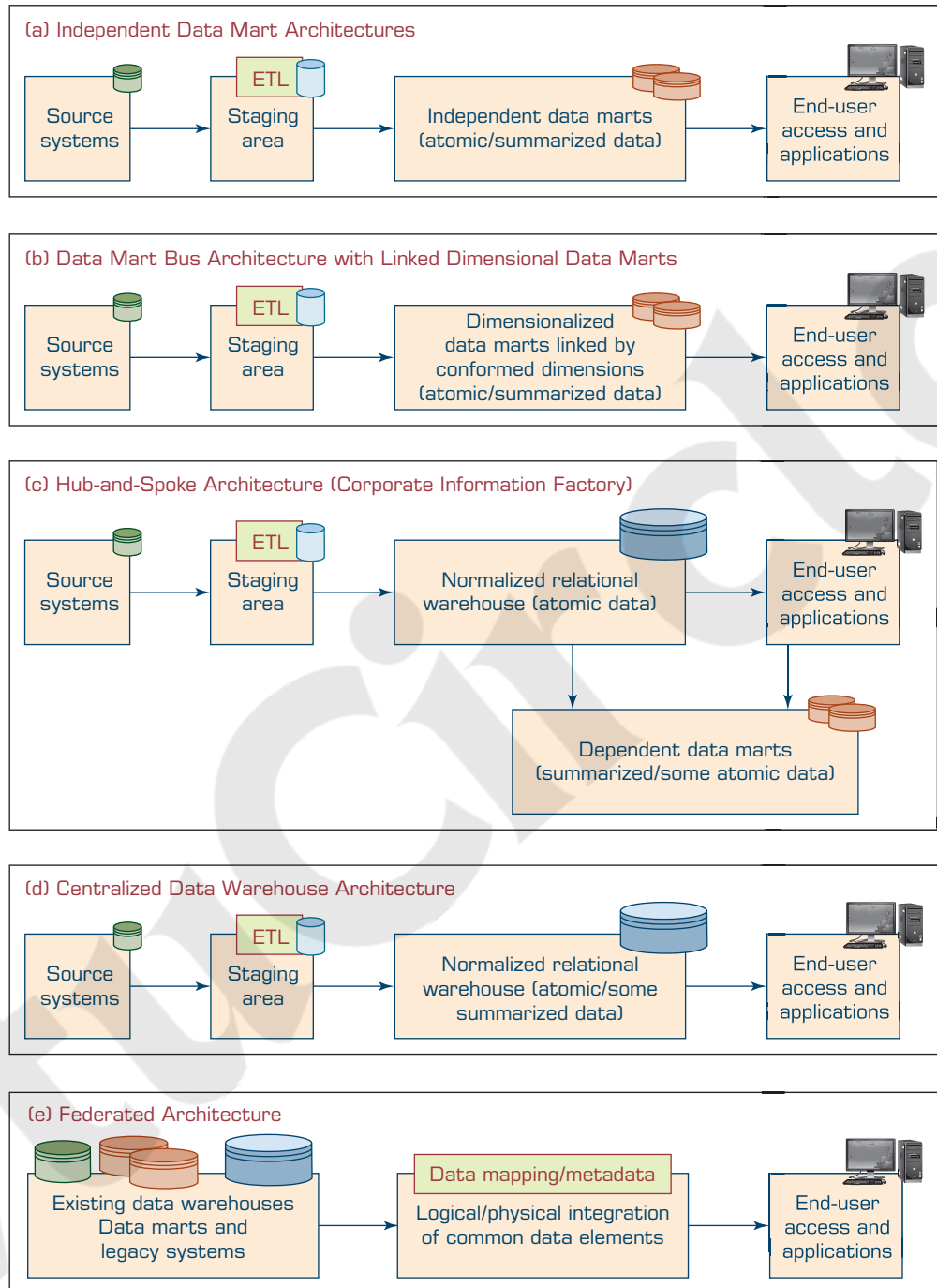


FIGURE 4.8 Alternative Data Warehouse Architectures. Source: Adapted from Ariyachandra, T., & Watson, H. (2006b) Which data warehouse architecture is most successful? *Business Intelligence Journal*, 11(1), 4–6.

they may have inconsistent data definitions and different dimensions and measures, making it difficult to analyze data across the DMs (i.e., it is difficult, if not impossible, to get to the “one version of the truth”).

- **Data mart bus architecture.** This architecture is a viable alternative to the independent DMs where the individual marts are linked to each other via some

kind of middleware. Because the data are linked among the individual marts, there is a better chance of maintaining data consistency across the enterprise (at least at the metadata level). Even though it allows for complex data queries across DMs, the performance of these types of analysis may not be at a satisfactory level.

- **Hub-and-spoke architecture.** This is perhaps the most famous data warehousing architecture today. Here the attention is focused on building a scalable and maintainable infrastructure (often developed in an iterative way, subject area by subject area) that includes a centralized data warehouse and several dependent DMs (each for an organizational unit). This architecture allows for easy customization of user interfaces and reports. On the negative side, this architecture lacks the holistic enterprise view and may lead to data redundancy and data latency.
- **Centralized data warehouse.** The centralized data warehouse architecture is similar to the hub-and-spoke architecture except that there are no dependent DMs; instead, there is a gigantic EDW that serves the needs of all organizational units. This centralized approach provides users with access to all data in the data warehouse instead of limiting them to DMs. In addition, it reduces the amount of data the technical team has to transfer or change, therefore simplifying data management and administration. If designed and implemented properly, this architecture provides a timely and holistic view of the enterprise to whoever, whenever, and wherever they may be within the organization.
- **Federated data warehouse.** The federated approach is a concession to the natural forces that undermine the best plans for developing a perfect system. It uses all possible means to integrate analytical resources from multiple sources to meet changing needs or business conditions. Essentially, the federated approach involves integrating disparate systems. In a federated architecture, existing decision support structures are left in place, and data are accessed from those sources as needed. The federated approach is supported by middleware vendors that propose distributed query and join capabilities. These eXtensible Markup Language (XML)-based tools offer users a global view of distributed data sources, including data warehouses, DMs, Web sites, documents, and operational systems. When users choose query objects from this view and press the submit button, the tool automatically queries the distributed sources, joins the results, and presents them to the user. Because of performance and data quality issues, most experts agree that federated approaches work well to supplement data warehouses, not replace them (see Eckerson, 2005).

Following are the top 10 factors that potentially affect the architecture selection decision:

1. Information interdependence between organizational units
2. Upper management's information needs
3. Urgency of need for a data warehouse
4. Nature of end-user tasks
5. Constraints on resources
6. Strategic view of the data warehouse prior to implementation
7. Compatibility with existing systems
8. Perceived ability of the in-house IT staff
9. Technical issues
10. Social/political factors

These factors are similar to many success factors described in the literature for information system projects and DSS and BI projects. Technical issues, beyond providing technology that is feasibly ready for use, are important, but often not as important as behavioral issues, such as meeting upper management's information needs and user

involvement in the development process (a social/political factor). Each data warehousing architecture has specific applications for which it is most (and least) effective and thus provides maximal benefits to the organization. However, overall, the DM structure seems to be the least effective in practice. See Ariyachandra and Watson (2006a) for some additional details.

Which Architecture Is the Best?

Ever since data warehousing became a critical part of modern enterprises, the question of which data warehouse architecture is the best has been a topic of regular discussion. The two gurus of the data warehousing field, Bill Inmon and Ralph Kimball, are at the heart of this discussion. Inmon advocates the hub-and-spoke architecture (e.g., the Corporate Information Factory), whereas Kimball promotes the DM bus architecture with conformed dimensions. Other architectures are possible, but these two options are fundamentally different approaches, and each has strong advocates. To shed light on this controversial question, Ariyachandra and Watson (2006b) conducted an empirical study. To collect the data, they used a Web-based survey targeted at individuals involved in data warehouse implementations. Their survey included questions about the respondent, the respondent's company, the company's data warehouse, and the success of the data warehouse architecture.

In total, 454 respondents provided usable information. Surveyed companies ranged from small (less than \$10 million in revenue) to large (in excess of \$10 billion). Most of the companies were located in the United States (60%) and represented a variety of industries, with the financial services industry (15%) providing the most responses. The predominant architecture was the hub-and-spoke architecture (39%), followed by the bus architecture (26%), the centralized architecture (17%), independent DMs (12%), and the federated architecture (4%). The most common platform for hosting the data warehouses was Oracle (41%), followed by Microsoft (19%) and IBM (18%). The average (mean) gross revenue varied from \$3.7 billion for independent DMs to \$6 billion for the federated architecture.

They used four measures to assess the success of the architectures: (1) information quality, (2) system quality, (3) individual impacts, and (4) organizational impacts. The questions used a 7-point scale, with the higher score indicating a more successful architecture. Table 4.1 shows the average scores for the measures across the architectures.

TABLE 4.1 Average Assessment Scores for the Success of the Architectures

	Independent DMs	Bus Architecture	Hub-and-Spoke Architecture	Centralized Architecture (No Dependent DMs)	Federated Architecture
Information Quality	4.42	5.16	5.35	5.23	4.73
System Quality	4.59	5.60	5.56	5.41	4.69
Individual Impacts	5.08	5.80	5.62	5.64	5.15
Organizational Impacts	4.66	5.34	5.24	5.30	4.77

As the results of the study indicate, independent DMs scored the lowest on all measures. This finding confirms the conventional wisdom that independent DMs are a poor architectural solution. Next lowest on all measures was the federated architecture. Firms sometimes have disparate decision-support platforms resulting from mergers and acquisitions, and they may choose a federated approach, at least in the short term. The findings suggest that the federated architecture is not an optimal long-term solution. What is interesting, however, is the similarity of the averages for the bus, hub-and-spoke, and centralized architectures. The differences are sufficiently small that no claims can be made for a particular architecture's superiority over the others, at least based on a simple comparison of these success measures.

They also collected data on the domain (e.g., varying from a subunit to company-wide) and the size (i.e., amount of data stored) of the warehouses. They found that the hub-and-spoke architecture is typically used with more enterprise-wide implementations and larger warehouses. They also investigated the cost and time required to implement the different architectures. Overall, the hub-and-spoke architecture was the most expensive and time-consuming to implement.

► SECTION 4.4 REVIEW QUESTIONS

1. What are the key similarities and differences between a two-tiered architecture and a three-tiered architecture?
2. How has the Web influenced data warehouse design?
3. List the alternative data warehousing architectures discussed in this section.
4. What issues should be considered when deciding which architecture to use in developing a data warehouse? List the 10 most important factors.
5. Which data warehousing architecture is the best? Why?

4.5 DATA MANAGEMENT AND WAREHOUSE DEVELOPMENT

A data warehousing project is a major undertaking for any organization and is more complicated than a simple mainframe selection and implementation project because it comprises and influences many departments and many input and output interfaces and it can be part of a CRM business strategy. A data warehouse provides several benefits that can be classified as direct and indirect. Direct benefits include the following:

- End users can perform extensive analysis in numerous ways.
- A consolidated view of corporate data (i.e., a single version of the truth) is possible.
- Better and more timely information is possible. A data warehouse permits information processing to be relieved from costly operational systems onto low-cost servers; therefore, many more end-user information requests can be processed more quickly.
- Enhanced system performance can result. A data warehouse frees production processing because some operational system reporting requirements are moved to DSS.
- Data access is simplified.

Indirect benefits result from end users using these direct benefits. On the whole, these benefits enhance business knowledge, present a competitive advantage, improve customer service and satisfaction, facilitate decision making, and help in reforming business processes; therefore, they are the strongest contributions to competitive advantage (Parzinger & Frolick, 2001). For a detailed discussion of how organizations can obtain exceptional levels of payoffs, see Watson, Goodhue, and Wixom (2002). Given the potential benefits that a data warehouse can provide and the substantial investments

in time and money that such a project requires, it is critical that an organization structure its data warehouse project to maximize the chances of success. In addition, the organization must, obviously, take costs into consideration. Kelly (2001) described an ROI approach that considers benefits in the categories of keepers (i.e., money saved by improving traditional decision support functions), gatherers (i.e., money saved due to automated collection and dissemination of information), and users (i.e., money saved or gained from decisions made using the data warehouse). Costs include those related to hardware, software, network bandwidth, internal development, internal support, training, and external consulting. The net present value is calculated over the expected life of the data warehouse. Because the benefits are broken down approximately as 20% for keepers, 30% for gatherers, and 50% for users, Kelly indicated that users should be involved in the development process, a success factor typically mentioned as critical for systems that imply change in an organization.

Clearly defining the business objective, gathering project support from management end users, setting reasonable time frames and budgets, and managing expectations are critical to a successful data warehousing project. A data warehousing strategy is a blueprint for the successful introduction of the data warehouse. The strategy should describe where the company wants to go, why it wants to go there, and what it will do when it gets there. It needs to take into consideration the organization's vision, structure, and culture. See Matney (2003) for the steps that can help in developing a flexible and efficient support strategy. When the plan and support for a data warehouse are established, the organization needs to examine data warehouse vendors. (See Table 4.2 for a sample list of vendors; also see The Data Warehousing Institute [twidi.org] and Information Builders [informationbuilders.com].) Many vendors provide software demos of their data warehousing and BI products.

Data Warehouse Development Approaches

Many organizations need to create the data warehouses used for decision support. Two competing approaches are employed. The first approach is that of Bill Inmon, who is often called “the father of data warehousing.” Inmon supports a top-down development approach that adapts traditional relational database tools to the development needs of an enterprise-wide data warehouse, also known as the EDW approach. The second approach is that of Ralph Kimball, who proposed a bottom-up approach that employs dimensional modeling, also known as the DM approach. Knowing how these two models are alike and how they differ helps us understand the basic data warehouse concepts. Table 4.3 compares the two approaches. We describe these approaches in detail next.

THE INMON MODEL: THE EDW APPROACH Inmon's approach emphasizes top-down development, employing established database development methodologies and tools, such as entity-relationship diagrams (ERD) and an adjustment of the spiral development approach. The EDW approach does not preclude the creation of DMs. The EDW is the ideal in this approach because it provides a consistent and comprehensive view of the enterprise. Murtaza (1998) presented a framework for developing EDW.

THE KIMBALL MODEL: THE DATA MART APPROACH Kimball's DM strategy is a “plan big, build small” approach. A DM is a subject-oriented or department-oriented data warehouse. It is a scaled-down version of a data warehouse that focuses on the requests of a specific department, such as marketing or sales. This model applies dimensional data modeling, which starts with tables. Kimball advocated a development methodology that entails a bottom-up approach, which in the case of data warehouses means building one DM at a time.

TABLE 4.2 Sample List of Data Warehousing Vendors

Vendor	Product Offerings
Business Objects (businessobjects.com)	A Comprehensive set of BI and data visualization software (now owned by SAP)
Computer Associates (cai.com)	Comprehensive set of data warehouse (DW) tools and products
DataMirror (datamirror.com)	DW administration, management, and performance products
Data Advantage Group (dataadvantagegroup.com)	Metadata software
Dell (dell.com)	DW servers
Embarcadero Technologies (embarcadero.com)	DW administration, management, and performance products
Greenplum (greenplum.com)	Data warehousing and data appliance solution provider (now owned by EMC)
Harte-Hanks (harte-hanks.com)	Customer relationship management (CRM) products and services
HP (hp.com)	DW servers
Hummingbird Ltd. (hummingbird.com)	DW engines and exploration warehouses
Hyperion Solutions (hyperion.com)	Comprehensive set of DW tools, products, and applications
IBM InfoSphere (www-01.ibm.com/software/data/infosphere)	Data integration, DW, master data management, Big Data products
Informatica (informatica.com)	DW administration, management, and performance products
Microsoft (microsoft.com)	DW tools and products
Netezza	DW software and hardware (DW appliance) provider (now owned by IBM)
Oracle (including PeopleSoft and Siebel; oracle.com)	DW, ERP, and CRM tools, products, and applications
SAS Institute (sas.com)	DW tools, products, and applications
Siemens (siemens.com)	DW servers
Sybase (sybase.com)	Comprehensive set of DW tools and applications
Teradata (teradata.com)	DW tools, DW appliances, DW consultancy, and applications

WHICH MODEL IS BEST? There is no one-size-fits-all strategy to data warehousing. An enterprise's data warehousing strategy can evolve from a simple DM to a complex data warehouse in response to user demands, the enterprise's business requirements, and the enterprise's maturity in managing its data resources. For many enterprises, a DM is frequently a convenient first step to acquiring experience in constructing and managing a data warehouse while presenting business users with the benefits of better access to their data; in addition, a DM commonly indicates the business value of data warehousing. Ultimately, engineering an EDW that consolidates old DMs and data warehouses is the ideal solution.

TABLE 4.3 Contrasts between the DM and EDW Development Approaches

Effort	DM Approach	EDW Approach
Scope	One subject area	Several subject areas
Development time	Months	Years
Development cost	\$10,000 to \$100,000+	\$1,000,000+
Development difficulty	Low to medium	High
Data prerequisite for sharing	Common (within business area)	Common (across enterprise)
Sources	Only some operational and external systems	Many operational and external systems
Size	Megabytes to several gigabytes	Gigabytes to petabytes
Time horizon	Near-current and historical data	Historical data
Data transformations	Low to medium	High
Update frequency	Hourly, daily, weekly	Weekly, monthly
Technology		
Hardware	Workstations and departmental servers	Enterprise servers and mainframe computers
Operating system	Windows and Linux	Unix, Z/OS, OS/390
Databases	Workgroup or standard database servers	Enterprise database servers
Usage		
Number of simultaneous users	10s	100s to 1,000s
User types	Business area analysts and managers	Enterprise analysts and senior executives
Business spotlight	Optimizing activities within the business area	Crossfunctional optimization and decision making

Sources: Adapted from Van den Hoven, J. (2003). Data marts: Plan big, build small. In *IS Management Handbook*, 8th ed., Boca Raton, FL: CRC Press; Ariyachandra, T., & Watson, H. (2006b). Which data warehouse architecture is most successful? *Business Intelligence Journal*, 11(1), 4–6.

However, the development of individual DMs can often provide many benefits along the way toward developing an EDW, especially if the organization is unable or unwilling to invest in a large-scale project. DMs can also demonstrate feasibility and success in providing benefits. This could potentially lead to an investment in an EDW. Table 4.4 summarizes the most essential characteristic differences between the two models.

Additional Data Warehouse Development Considerations

Some organizations want to completely outsource their data warehousing efforts. They simply do not want to deal with software and hardware acquisitions, and they do not want to manage their information systems. One alternative is to use hosted data warehouses. In this scenario, another firm—ideally, one that has a lot of experience and expertise—develops and maintains the data warehouse. However, there are security and privacy concerns with this approach. See Technology Insights 4.1 for some details.

TABLE 4.4 Essential Differences between Inmon's and Kimball's Approaches

Characteristic	Inmon	Kimball
<i>Methodology and Architecture</i>		
Overall approach	Top-down	Bottom-up
Architecture structure	Enterprise-wide (atomic) data warehouse "feeds" departmental database	DMs model a single business process, and enterprise consistency is achieved through a data bus and conformed dimensions
Complexity of the method	Quite complex	Fairly simple
Comparison with established development methodologies	Derived from the spiral methodology	Four-step process; a departure from RDBMS methods
Discussion of physical design	Fairly thorough	Fairly light
<i>Data Modeling</i>		
Data orientation	Subject- or data-driven	Process-oriented
Tools	Traditional (entity-relationship diagrams [ERD], data flow diagrams [DFD])	Dimensional modeling; a departure from relational modeling
End-user accessibility	Low	High
<i>Philosophy</i>		
Primary audience	IT professionals	End users
Place in the organization	Integral part of the corporate information factory	Transformer and retainer of operational data
Objective	Deliver a sound technical solution based on proven database methods and technology	Deliver a solution that makes it easy for end users to directly query the data and still get reasonable response times

Sources: Adapted from Breslin, M. (2004, Winter). Data warehousing battle of the giants: Comparing the basics of Kimball and Inmon models. *Business Intelligence Journal*, 9(1), 6–20; Ariyachandra, T., & Watson, H. (2006b). Which data warehouse architecture is most successful? *Business Intelligence Journal*, 11(1).

TECHNOLOGY INSIGHTS 4.1 Hosted Data Warehouses

A hosted data warehouse has nearly the same, if not more, functionality as an on-site data warehouse, but it does not consume computer resources on client premises. A hosted data warehouse offers the benefits of BI minus the cost of computer upgrades, network upgrades, software licenses, in-house development, and in-house support and maintenance.

A hosted data warehouse offers the following benefits:

- Requires minimal investment in infrastructure
- Frees up capacity on in-house systems
- Frees up cash flow
- Makes powerful solutions affordable
- Enables powerful solutions that provide for growth
- Offers better-quality equipment and software

- Provides faster connections
- Enables users to access data from remote locations
- Allows a company to focus on core business
- Meets storage needs for large volumes of data

Despite its benefits, a hosted data warehouse is not necessarily a good fit for every organization. Large companies with revenue upwards of \$500 million could lose money if they already have underused internal infrastructure and IT staff. Furthermore, companies that see the paradigm shift of outsourcing applications as loss of control of their data are not likely to use a BI service provider. Finally, the most significant and common argument against implementing a hosted data warehouse is that it may be unwise to outsource sensitive applications for reasons of security and privacy.

Sources: Compiled from Thornton, M., & Lampa, M. (2002). Hosted data warehouse. *Journal of Data Warehousing*, 7(2), 27–34; Thornton, M. (2002, March 18). What about security? The most common, but unwarranted, objection to hosted data warehouses. *DM Review*, 12(3), 30–43.

Representation of Data in Data Warehouse

A typical data warehouse structure is shown in Figure 4.4. Many variations of data warehouse architecture are possible (see Figure 4.8). No matter what the architecture was, the design of data representation in the data warehouse has always been based on the concept of dimensional modeling. **Dimensional modeling** is a retrieval-based system that supports high-volume query access. Representation and storage of data in a data warehouse should be designed in such a way that not only accommodates but also boosts the processing of complex multidimensional queries. Often, the star schema and the snowflake schema are the means by which dimensional modeling is implemented in data warehouses.

The **star schema** (sometimes referenced as star join schema) is the most commonly used and the simplest style of dimensional modeling. A star schema contains a central fact table surrounded by and connected to several **dimension tables** (Adamson, 2009). The fact table contains a large number of rows that correspond to observed facts and external links (i.e., foreign keys). A fact table contains the descriptive attributes needed to perform decision analysis and query reporting, and foreign keys are used to link to dimension tables. The decision analysis attributes consist of performance measures, operational metrics, aggregated measures (e.g., sales volumes, customer retention rates, profit margins, production costs, scrap rate), and all the other metrics needed to analyze the organization's performance. In other words, the fact table primarily addresses what the data warehouse supports for decision analysis.

Surrounding the central fact tables (and linked via foreign keys) are dimension tables. The dimension tables contain classification and aggregation information about the central fact rows. Dimension tables contain attributes that describe the data contained within the fact table; they address how data will be analyzed and summarized. Dimension tables have a one-to-many relationship with rows in the central fact table. In querying, the dimensions are used to slice and dice the numerical values in the fact table to address the requirements of an ad hoc information need. The star schema is designed to provide fast query-response time, simplicity, and ease of maintenance for read-only database structures. A simple star schema is shown in Figure 4.9a. The star schema is considered a special case of the snowflake schema.

The **snowflake schema** is a logical arrangement of tables in a multidimensional database in such a way that the entity-relationship diagram resembles a snowflake in shape. Closely related to the star schema, the snowflake schema is represented by centralized fact tables (usually only one), which are connected to multiple dimensions. In the snowflake schema, however, dimensions are normalized into multiple

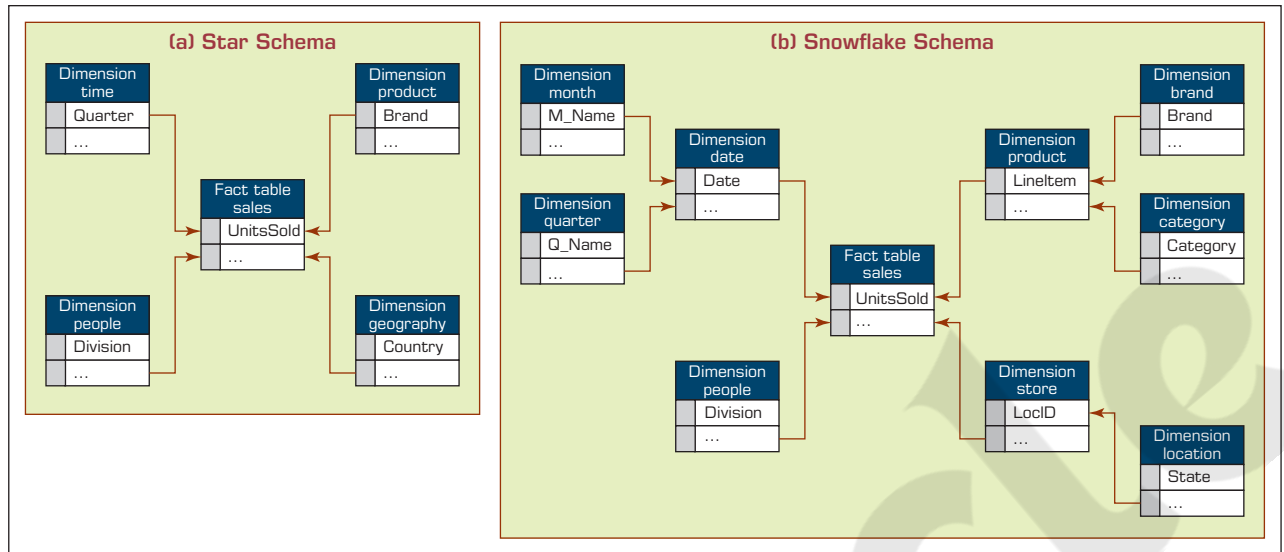


FIGURE 4.9 (a) The Star Schema, and (b) the Snowflake Schema.

related tables, whereas the star schema's dimensions are denormalized, with each dimension being represented by a single table. A simple snowflake schema is shown in Figure 4.9b.

Analysis of Data in Data Warehouse

Once the data is properly stored in a data warehouse, it can be used in various ways to support organizational decision making. OLAP is arguably the most commonly used data analysis technique in data warehouses, and it has been growing in popularity due to the exponential increase in data volumes and the recognition of the business value of data-driven analytics. Simply, OLAP is an approach to quickly answer ad hoc questions by executing multidimensional analytical queries against organizational data repositories (i.e., data warehouses, DMs).

OLAP versus OLTP

OLTP (online transaction processing system) is a term used for a transaction system that is primarily responsible for capturing and storing data related to day-to-day business functions such as ERP, CRM, SCM, POS, and so forth. An OLTP system addresses a critical business need, automating daily business transactions, and running real-time reports and routine analysis. But these systems are not designed for ad hoc analysis and complex queries that deal with a number of data items. OLAP, on the other hand, is designed to address this need by providing ad hoc analysis of organizational data much more effectively and efficiently. OLAP and OLTP rely heavily on each other: OLAP uses the data captured by OLTP, and OLTP automates the business processes that are managed by decisions supported by OLAP. Table 4.5 provides a multicriteria comparison between OLTP and OLAP.

OLAP Operations

The main operational structure in OLAP is based on a concept called cube. A **cube** in OLAP is a multidimensional data structure (actual or virtual) that allows fast analysis of data. It can also be defined as the capability of efficiently manipulating and analyzing data from multiple perspectives. The arrangement of data into cubes aims to overcome a limitation of relational databases: Relational databases are not well suited for

TABLE 4.5 A Comparison between OLTP and OLAP

Criteria	OLTP	OLAP
Purpose	To carry out day-to-day business functions	To support decision making and provide answers to business and management queries
Data source	Transaction database (a normalized data repository primarily focused on efficiency and consistency)	Data warehouse or DM (a nonnormalized data repository primarily focused on accuracy and completeness)
Reporting	Routine, periodic, narrowly focused reports	Ad hoc, multidimensional, broadly focused reports and queries
Resource requirements	Ordinary relational database	Multiprocessor, large-capacity specialized database
Execution speed	Fast (recording of business transaction and routine reports)	Slow (resource intensive, complex, large-scale queries)

near instantaneous analysis of large amounts of data. Instead, they are better suited for manipulating records (adding, deleting, and updating data) that represent a series of transactions. Although many report-writing tools exist for relational databases, these tools are slow when a multidimensional query that encompasses many database tables needs to be executed.

Using OLAP, an analyst can navigate through the database and screen for a particular subset of the data (and its progression over time) by changing the data's orientations and defining analytical calculations. These types of user-initiated navigation of data through the specification of slices (via rotations) and **drill down/up** (via aggregation and disaggregation) is sometimes called "slice and dice." Commonly used OLAP operations include slice and dice, drill down, roll-up, and pivot.

- **Slice.** A slice is a subset of a multidimensional array (usually a two-dimensional representation) corresponding to a single value set for one (or more) of the dimensions not in the subset. A simple slicing operation on a three-dimensional cube is shown in Figure 4.10.
- **Dice.** The dice operation is a slice on more than two dimensions of a data cube.
- **Drill Down/Up.** Drilling down or up is a specific OLAP technique whereby the user navigates among levels of data ranging from the most summarized (up) to the most detailed (down).
- **Roll-up.** A roll-up involves computing all the data relationships for one or more dimensions. To do this, a computational relationship or formula might be defined.
- **Pivot.** This is used to change the dimensional orientation of a report or ad hoc query-page display.

Data Integration and the Extraction, Transformation, and Load (ETL) Processes

Global competitive pressures, demand for return on investment (ROI), management and investor inquiry, and government regulations are forcing business managers to rethink how they integrate and manage their businesses. A decision maker typically needs access to multiple sources of data that must be integrated. Before data warehouses, DMs, and BI software, providing access to data sources was a major, laborious process. Even with modern Web-based data management tools, recognizing what data to access and providing

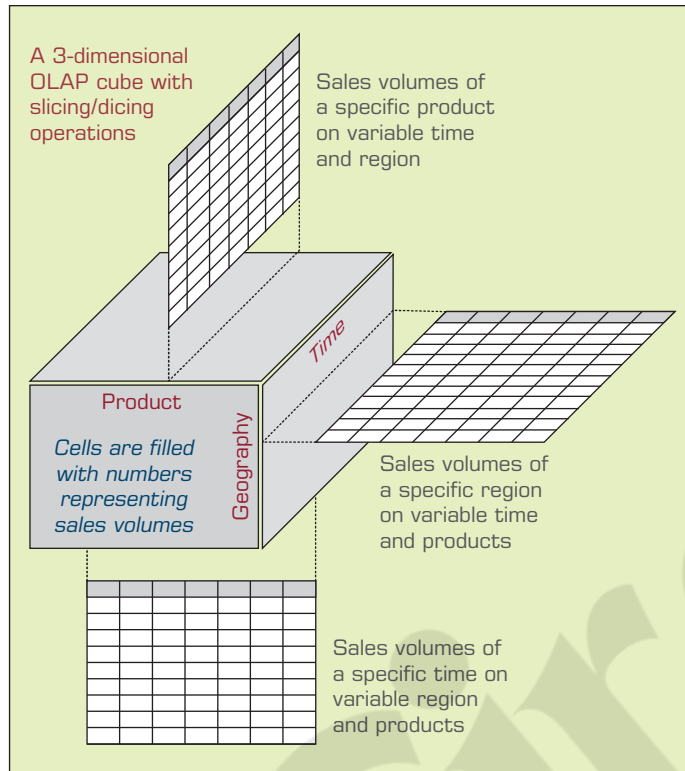


FIGURE 4.10 Slicing Operations on a Simple Three-Dimensional Data Cube.

them to the decision maker is a nontrivial task that requires database specialists. As data warehouses grow in size, the issues of integrating data grow as well.

The business analysis needs continue to evolve. Mergers and acquisitions, regulatory requirements, and the introduction of new channels can drive changes in BI requirements. In addition to historical, cleansed, consolidated, and point-in-time data, business users increasingly demand access to real-time, unstructured, and/or remote data. And everything must be integrated with the contents of an existing data warehouse. Moreover, access via PDAs and through speech recognition and synthesis is becoming more commonplace, further complicating integration issues (Edwards, 2003). Many integration projects involve enterprise-wide systems. Orovic (2003) provided a checklist of what works and what does not work when attempting such a project. Properly integrating data from various databases and other disparate sources is difficult. When it is not done properly, though, it can lead to disaster in enterprise-wide systems such as CRM, ERP, and supply-chain projects (Nash, 2002).

Data Integration

Data integration comprises three major processes that, when correctly implemented, permit data to be accessed and made accessible to an array of ETL and analysis tools and the data warehousing environment: data access (i.e., the ability to access and extract data from any data source), data federation (i.e., the integration of business views across multiple data stores), and change capture (based on the identification, capture, and delivery of the changes made to enterprise data sources). Analytics in Action 4.2 shows how a large nonprofit organization achieved extraordinary results by transforming and modernizing their BI infrastructure with modern-day data warehousing appliances.

ANALYTICS IN ACTION 4.2 AARP Transforms Its BI Infrastructure and Achieves a 347% ROI in Three Years

AARP, Inc., formerly the American Association of Retired Persons, is a U.S.-based membership and interest group, founded in 1958 by Ethel Percy Andrus, PhD, a retired educator from California, and Leonard Davis, founder of Colonial Penn Group of insurance companies. As described in their Web site (aarp.org), AARP is a nonprofit, nonpartisan, social welfare organization with a membership of nearly 38 million that helps people turn their goals and dreams into real possibilities, strengthens communities, and fights for the issues that matter most to families—such as healthcare, employment and income security, and protection from financial abuse.

A Growing Demand for BI

In 2002, the organization first launched a BI initiative that would centralize information (AARP has offices in all 50 states as well as the District of Columbia) and empower its staff with current, relevant, accurate, and flexible analytics to:

- Match services and product offerings to membership base and expectations.
- Improve member profitability, retention, and acquisition.
- Protect AARP brand image by managing relationships with third-party service providers.

This insight helped fuel AARP's success, and with this success came larger data volumes and an increased demand for new analytics.

By 2009, the BI team faced a new challenge. Its data warehouse—based on an SQL relational database from Oracle—could no longer keep up with the demand. The team experienced more than 30 system failures that year. This was both unacceptable and costly.

System performance was a key concern as well. As the data volumes grew, daily loads into the warehouse couldn't be completed until 3:00 P.M.—which affected how long staff had to wait for reports. “Our analysts would run a report, then go for coffee or for lunch, and, maybe if they were lucky, by 5:00 P.M. they would get the response,” says Bruni, Practice Director, Business Intelligence, AARP. “It was unacceptable. The system was so busy writing the new daily data that it didn't give any importance to the read operations performed by users.”

Analysts also couldn't create ad hoc queries without IT intervention. When IT received a request for a new type of report, the BI team would have to optimize the queries and send a report sample back to the requestors for review. The process, from start to finish, could take weeks to months. Finally, with more than 36 terabytes of data in the data warehouse, staff found it impossible to back up the system each night. Backups were limited to a few critical tables, making it difficult for staff to create an effective disaster recovery plan.

According to Bruni, if left unsolved, these challenges could have affected AARP's work. “Analytics provide key metrics that are critical to evaluate how well our membership and social goals are being attained,” says Bruni. “It is essential to enabling continuous improvement and decision making to support member needs.”

Creating an Agile BI Environment

As Bruni's team looked to modernize the BI environment, they evaluated two options—upgrading the existing environment or moving to a single data warehouse appliance. “We found the cost of each option comparable, but only the appliance provided us a paradigm shift in terms of the performance we needed,” says Bruni. “Among the different partners we looked at, the IBM Netezza data warehouse appliance provided the safest bet because it didn't require the data model fine-tuning that other data warehouses do. We were also able to try the solution before we bought it to see whether it really could do everything we needed. Most vendors do not provide this type of ‘try-before-you-buy’ option.”

In building the new environment, the organization adopted a “Scrum” development model, usually used by software developers, to provide a framework that shortens development cycles and speeds time to market for BI requests. “Using Scrum in data warehousing is kind of unheard of,” says Bruni. “But the basic premise it provides is an agile, iterative process that enables us to rapidly transform our users’ analytic needs into operating reports that show meaningful data.”

Within nine months from the acquisition of its new platform, the team had converted all the scripts and procedures from Oracle Database into the IBM® Netezza® data warehouse appliance. Core accounts and membership data (which resides on an IBM DB2® for z/OS® database running on an IBM System z® server), financial and human resource data from other smaller databases, and campaign analysis and segmentation data from third-party data sources are now loaded in the IBM Netezza data warehouse appliance nightly and accessible via the organization’s BI tools without interruption.

Running Complex Queries at Lightning Speed

In terms of performance (which was the BI team’s most pressing concern), daily data loads are now completed before 8:00 A.M.—a 1400% improvement—and reports that previously took minutes to run are completed in seconds—a 1700% improvement. The solution also helped compress the data size from 36 terabytes to just 1.5 terabytes, enabling staff to easily back up the data warehouse in only 30 minutes.

Equally important, the nearly 220 human resources, finance, marketing, and campaign staff members that use the system can now conduct what Bruni refers to as “train-of-thought analysis”—creating ad hoc reports to test theories regarding membership needs. “The IBM Netezza data warehouse appliance is like driving a Ferrari,” says Bruni. “We have opened a whole new realm of possibilities to our internal customers, who are actually able to create reports on-the-fly and get the results back in a matter of seconds. In the first few months of operation, we saw a huge spike in the number of reports being created—nearly three times the number that we had previously supported. With the deep dive they can conduct now, we’ve seen a steady growth in member renewals, acquisitions and engagement.”

Achieving Rapid ROI

The new platform has also enabled the organization to redeploy IT support staff from the BI group to other areas. Previously, the team needed one full-time database administrator (DBA) along with part-time support from the organization’s storage area network (SAN) and midrange service teams. “It’s amazing,” says Bruni. “We no longer need IT support. The IBM Netezza data warehouse appliance is shipped already optimized. Give it power, give it network, and you’re done. It doesn’t need anything else.”

These improvements have enabled the organization to realize a 9% return on investment in the first year, with an anticipated 274% ROI by the second year, and a 347% investment by the third year. “Our initial analysis projected a positive ROI already in the first year—which is very unusual for infrastructure upgrades given all costs are incurrent in the first year,” says Bruni. “Our actual ROI post-implementation was even higher as we completed the swap three months ahead of schedule.”

Expanding the Influence of BI

By modernizing its infrastructure, Bruni’s team has elevated the value and perception of BI in the organization. “After we moved to IBM Netezza, the word spread that we were doing things right and that leveraging us as an internal service was really smart,” says Bruni. “We’ve gained new mission-critical areas, such as the social-impact area which supports our Drive to End Hunger and Create the Good campaigns, based on the fact that we have such a robust infrastructure and that we changed our approach to business. We can develop in a more agile way from a development standpoint. From a program management standpoint, it shrinks our release cycles from months, which is typical with traditional data warehouse infrastructures, to just weeks.”

QUESTIONS FOR DISCUSSION

1. What were the challenges AARP was facing?
2. What was the approach for a potential solution?
3. What were the results obtained in the short term, and what were the future plans?

Source: IBM customer success story. (2021). “AARP transforms its business intelligence infrastructure—Achieving a 347% ROI in three years from BI modernization effort,” <http://www-03.ibm.com/software/businesscasestudies/us/en/corp?synkey=A735189Y23828M82> (accessed June 2021) and <https://www.merkle.com/thought-leadership/case-studies/aarpr-launches-digital-transformation-drive-personalization> (accessed May 2023).

A major purpose of a data warehouse is to integrate data from multiple systems. Various integration technologies enable data and metadata integration:

- Enterprise application integration (EAI)
- Service-oriented architecture (SOA)
- Enterprise information integration (EII)
- Extraction, transformation, and load (ETL)

Enterprise application integration (EAI) provides a vehicle for pushing data from source systems into the data warehouse. It involves integrating application functionality and is focused on sharing functionality (rather than data) across systems, thereby enabling flexibility and reuse. Traditionally, EAI solutions have focused on enabling application reuse at the application programming interface level. Recently, EAI is accomplished by using SOA coarse-grained services (a collection of business processes or functions) that are well defined and documented. Using Web services is a specialized way of implementing an SOA. EAI can be used to facilitate data acquisition directly into a near-real-time data warehouse or to deliver decisions to the OLTP systems. There are many different approaches to and tools for EAI implementation.

Enterprise information integration (EII) is an evolving tool space that promises real-time data integration from a variety of sources, such as relational databases, Web services, and multidimensional databases. It is a mechanism for pulling data from source systems to satisfy a request for information. EII tools use predefined metadata to populate views that make integrated data appear relational to end users. XML may be the most important aspect of EII because XML allows data to be tagged either at creation time or later. These tags can be extended and modified to accommodate almost any area of knowledge (see Kay, 2005).

Physical data integration has conventionally been the main mechanism for creating an integrated view with data warehouses and DMs. With the advent of EII tools (see Kay, 2005), new virtual data integration patterns are feasible. Manglik and Mehra (2005) discussed the benefits and constraints of new data integration patterns that can expand traditional physical methodologies to present a comprehensive view for the enterprise.

We next turn to the approach for loading data into the warehouse: ETL.

Extraction, Transformation, and Load

At the heart of the technical side of the data warehousing process is **extraction, transformation, and load (ETL)**. ETL technologies, which have existed for some time, are instrumental in the process and use of data warehouses. The ETL process is an integral component in any data-centric project. IT managers are often faced with challenges because the ETL process typically consumes 70% of the time in a data-centric project.

The ETL process consists of extraction (i.e., reading data from one or more databases), transformation (i.e., converting the extracted data from its previous form into the form in which it needs to be so that it can be placed into a data warehouse or simply another database), and load (i.e., putting the data into the data warehouse). Transformation occurs by using rules or lookup tables or by combining the data with other data. The three database functions are integrated into one tool to pull data out of one or more databases and place them into another, consolidated database or a data warehouse.

ETL tools also transport data between sources and targets, document how data elements (e.g., metadata) change as they move between source and target, exchange metadata with other applications as needed, and administer all runtime processes and operations (e.g., scheduling, error management, audit logs, statistics). ETL is extremely important for data integration as well as for data warehousing. The purpose of the ETL process is to load the warehouse with integrated and cleansed data. The data used in ETL processes can come from any source: a mainframe application, an ERP application, a CRM tool, a flat file, an Excel spreadsheet, or even a message queue. In Figure 4.11, we outline the ETL process.

The process of migrating data to a data warehouse involves the extraction of data from all relevant sources. Data sources may consist of files extracted from OLTP databases, spreadsheets, personal databases (e.g., Microsoft Access), or external files. Typically, all the input files are written to a set of staging tables, which are designed to facilitate the load process. A data warehouse contains numerous business rules that define such things as how the data will be used, summarization rules, standardization of encoded attributes, and calculation rules. Any data quality issues pertaining to the source files need to be corrected before the data are loaded into the data warehouse. One of the benefits of a well-designed data warehouse is that these rules can be stored in a metadata repository and applied to the data warehouse centrally. This differs from an OLTP approach, which typically has data and business rules scattered throughout the system. The process of loading data into a data warehouse can be performed either through data transformation tools that provide a GUI to aid in the development and maintenance of business rules or through more traditional methods, such as developing programs or utilities to load the data warehouse, using programming languages such as PL/SQL, C++, Java, or .NET Framework languages. This decision is not easy for organizations. Several issues affect

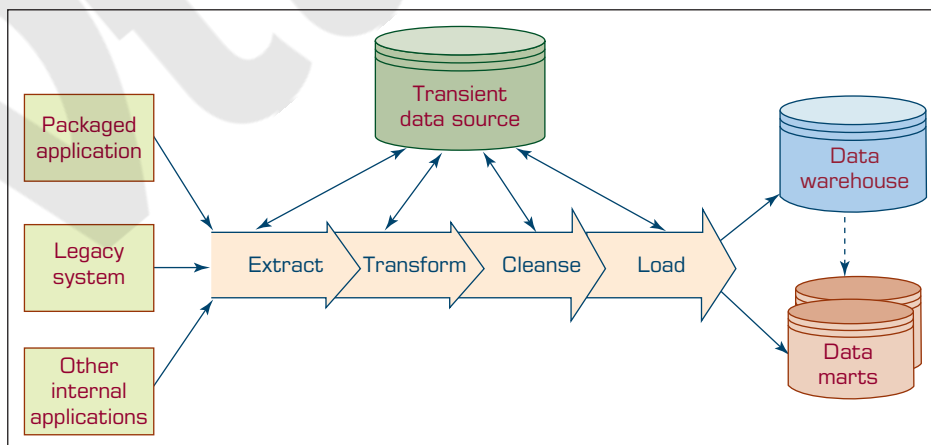


FIGURE 4.11 The ETL Process.

whether an organization will purchase data transformation tools or build the transformation process itself:

- Data transformation tools are expensive.
- Data transformation tools may have a long learning curve.
- It is difficult to measure how the IT organization is doing until it has learned to use the data transformation tools.

In the long run, a transformation-tool approach should simplify the maintenance of an organization's data warehouse. Transformation tools can also be effective in detecting and scrubbing (i.e., removing any anomalies in the data). OLAP and data mining tools rely on how well the data are transformed.

As an example of effective ETL, Motorola, Inc., uses ETL to feed its data warehouses. Motorola collects information from 30 different procurement systems and sends them to its global SCM data warehouse for analysis of aggregate company spending (see Songini, 2004).

Solomon (2005) classified ETL technologies into four categories: sophisticated, enabler, simple, and rudimentary. It is generally acknowledged that tools in the sophisticated category will result in the ETL process being better documented and more accurately managed as the data warehouse project evolves.

Even though it is possible for programmers to develop software for ETL, it is simpler to use an existing ETL tool. The following are some of the important criteria in selecting an ETL tool (see Brown, 2004):

- Ability to read from and write to an unlimited number of data source architectures
- Automatic capturing and delivery of metadata
- A history of conforming to open standards
- An easy-to-use interface for the developer and the functional user

Performing extensive ETL may be a sign of poorly managed data and a fundamental lack of a coherent data management strategy. There is a direct correlation between the extent of redundant and inconsistent data and the number of ETL tasks performed in the data pre-processes. When data are managed correctly as an enterprise asset, ETL efforts are significantly reduced, and redundant data are eliminated. This leads to huge savings in data maintenance and greater efficiency in new developments while also improving data quality. Poorly designed ETL processes are costly to maintain, change, and update. Consequently, it is crucial to make the proper choices in terms of the technology and tools to use for developing and maintaining the ETL process.

Numerous packaged ETL tools are available in the market. Database and data warehouse vendors currently offer ETL capabilities that both enhance and compete with independent ETL tools. SAS acknowledges the importance of data quality and offers the industry's first fully integrated solution that merges ETL and data quality to transform data into strategic valuable assets. Other integrated ETL software providers include Microsoft, Oracle, IBM, Informatica, and Salesforce.

► SECTION 4.5 REVIEW QUESTIONS

1. List the benefits of data warehouses.
2. List several criteria for selecting a data warehouse vendor, and describe why they are important.
3. What is OLAP, and how does it differ from OLTP?
4. What is a cube? What do drill down, roll-up, slice, and dice mean?
5. Describe data integration.
6. Describe the three steps of the ETL process.
7. Why is the ETL process so important for data warehousing efforts?

4.6 DATA WAREHOUSE ADMINISTRATION, SECURITY ISSUES, AND FUTURE TRENDS

Data warehouses provide a distinct competitive edge to enterprises that effectively create and use them. Due to its huge size and its intrinsic nature, a data warehouse requires especially strong monitoring to sustain satisfactory efficiency and productivity. The successful administration and management of a data warehouse entails skills and proficiency that go past what is required of a traditional database administrator (DBA). A **data warehouse administrator (DWA)** should be familiar with high-performance software, hardware, and networking technologies. He or she should also possess solid business insight. Because data warehouses feed BI systems and DSS that help managers with their decision-making activities, the DWA should be familiar with the decision-making processes to suitably design and maintain the data warehouse structure. It is particularly significant for a DWA to keep the existing requirements and capabilities of the data warehouse stable while simultaneously providing flexibility for rapid improvements. Finally, a DWA must possess excellent communication skills. See Benander, Benander, Fadlalla, and James (2000) for a description of the key differences between a DBA and a DWA.

Security and privacy of information are main and significant concerns for a data warehouse professional. The U.S. government has passed regulations (e.g., the Gramm-Leach-Bliley privacy and safeguards rules, the Health Insurance Portability and Accountability Act of 1996 [HIPAA]), instituting obligatory requirements in the management of customer information. Hence, companies must create security procedures that are effective yet flexible to conform to numerous privacy regulations. According to Elson and LeClerc (2005), effective security in a data warehouse should focus on four main areas:

- Establishing effective corporate and security policies and procedures. An effective security policy should start at the top, with executive management, and should be communicated to all individuals within the organization.
- Implementing logical security procedures and techniques to restrict access. This includes user authentication, access controls, and encryption technology.
- Limiting physical access to the data center environment.
- Establishing an effective internal control review process with an emphasis on security and privacy.

In the near term, data warehousing developments will be determined by noticeable factors (e.g., data volumes, increased intolerance for latency, the diversity and complexity of data types) and less noticeable factors (e.g., unmet end-user requirements for dashboards, balanced scorecards, master data management, information quality). Given these drivers, Moseley (2009) and Agosta (2006) suggested that data warehousing trends will lean toward simplicity, value, and performance.

The Future of Data Warehousing

The field of data warehousing has been a vibrant area in IT in the last couple of decades, and the evidence in the BI/BA and Big Data world shows that the importance of the field will only get even more interesting. Following are some of the recently popularized concepts and technologies that will play a significant role in defining the future of data warehousing.

Sourcing (mechanisms for acquisition of data from diverse and dispersed sources):

- **Web, social media, and Big Data.** The recent upsurge in the use of the Web for personal as well as business purposes coupled with the tremendous interest in social media creates opportunities for analysts to tap into very rich data sources. Because of the sheer volume, velocity, and variety of the data, a new term, “Big

Data,” has been coined to name the phenomenon. Taking advantage of Big Data requires development of new and dramatically improved BI/BA technologies, which will result in a revolutionized data warehousing world.

- **Open source software.** Use of open source software tools is increasing at an unprecedented level in warehousing, BI, and data integration. There are good reasons for the upswing of open source software used in data warehousing (Russom, 2009): (1) the recession has driven up interest in low-cost open source software, (2) open source tools are coming into a new level of maturity, and (3) open source software augments traditional enterprise software without replacing it.
- **SaaS (software as a service), “The Extended ASP Model.”** SaaS is a creative way of deploying information systems applications where the provider licenses its applications to customers for use as a service on demand (usually over the Internet). SaaS software vendors may host the application on their own servers or upload the application to the consumer site. In essence, SaaS is the new and improved version of the ASP model. For data warehouse customers, finding SaaS-based software applications and resources that meet specific needs and requirements can be challenging. As these software offerings become more agile, the appeal and the actual use of SaaS as the choice of data warehousing platform will also increase.
- **Cloud computing.** Cloud computing is perhaps the newest and the most innovative platform choice to come along in years. Numerous hardware and software resources are pooled and virtualized, so that they can be freely allocated to applications and software platforms as resources are needed. This enables information systems applications to dynamically scale up as workloads increase. Although cloud computing and similar virtualization techniques are fairly well established for operational applications today, they are just now starting to be used as data warehouse platforms of choice. The dynamic allocation of a cloud is particularly useful when the data volume of the warehouse varies unpredictably, making capacity planning difficult.
- **Data lakes.** With the emergence of Big Data, there came a new data platform: data lake, which is a large storage location that can hold vast quantities of data (mostly unstructured) in its native/raw format for future/potential analytics consumption. Traditionally speaking, whereas a data warehouse stores structured data, a data lake stores all kinds of data. While they are both data storage mechanisms, a data warehouse is all about structured/tabular data and a data lake is about all types of data. Although much has been said and written about the relationship between the two (some of which suggests that data lake is the future name of data warehouses), as it stands, a data lake is not a replacement for a data warehouse; rather, they are complementary to one another. Inmon et al. (2021) provide some invaluable insight in developing successful data lake houses. Technology Insights 4.2 digs deeper into explaining data lakes and their role in the worlds of data warehousing and big data analytics.

TECHNOLOGY INSIGHTS 4.2 Data Lakes

With the emergence of Big Data phenomenon, a new term, “data lake,” has been coined. Many believe that a data lake is just the reincarnation of the good old data warehouse. The underlying assumption suggests that in the age of Big Data, the old way of data storage is not suitable (or sufficient) and therefore a new way of data storage/management is needed, which is paving the way for data lakes. Although most believe that a data lake is the way to go and are ready to jump in, others are standing back and being more cautious (and perhaps skeptical of its viability), calling it a swamp. So, what really is a data lake? Simply put, a data lake is a large storage location that can hold huge quantities of data (structured, unstructured, or semistructured) in its

native/raw format for a potential future use. Whereas a data warehouse stores structured data in related tables, files, or folders, a data lake uses a loosely defined (i.e., unstructured) architecture to store all kinds of data. The main commonality between a data lake and data warehouse is that they are both data storage mechanisms, and conversely, the main difference is that one is all about structured/tabular data and the other is about all kinds of data (i.e., Big Data).

Although the definition of data lake changes among data scientists, the most commonly used definition of it comes from James Dixon, the founder and CTO of Pentaho, who has also been credited with coming up with the term itself. This is how he describes a data lake (Dixon, 2010):

If you think of a data mart as a store of bottled water—cleansed and packaged and structured for easy consumption—the data lake is a large body of water in a more natural state. The contents of the data lake stream in from a source to fill the lake, and various users of the lake can come to examine, dive in, or take samples.

Perhaps the best way to characterize a data lake is to compare it to a data warehouse in a multidimensional table. Table 4.6 is a summary table (followed by brief descriptions) of the most commonly used dimensions to compare a data lake to a data warehouse (Dull, 2021; Campbell, 2015).

Data. A data warehouse only stores data that has been modeled/aggregated/structured, whereas a data lake stores all kinds of data—structured, semistructured, and unstructured—in its native/raw format.

Processing. Before loading data into a data warehouse, we first need to give it some shape and structure—that is, we need to model it into a star or snowflake schema, which is called schema-on-write. With a data lake, we just load in the raw data, as-is, and then when we are ready to use the data, we give it a shape or structure, which is called schema-on-read. These are two very different processing approaches.

Retrieval speed. For more than two decades, many algorithms have been developed to improve the speed at which the data is retrieved from large and feature-rich data warehouses. Such techniques included triggers, columnar data representation, in-database processing. As of now, the retrieval of data (which can be in any form or fashion—including unstructured text) is a time-demanding activity.

Storage. One of the primary features of Big Data technologies like Hadoop is that the cost of storing data is relatively low as compared to the data warehouse. There are two key reasons for this: First, Hadoop is open source software, so the licensing and community support is free. And second, Hadoop is designed to be installed on low-cost commodity hardware.

Agility. A data warehouse is a highly structured repository, by definition. It's not technically hard to change the structure, but it can be very time-consuming given all the business

TABLE 4.6 A Simple Comparison between a Data Warehouse and a Data Lake

Dimension	Data Warehouse	Data Lake
The nature of data	Structured, processed	Any data in raw/native format
Processing	Schema-on-write (SQL)	Schema-on-read (NoSQL)
Retrieval speed	Very fast	Slow
Cost	Expensive for large data volumes	Designed for low-cost storage
Agility	Less agile, fixed configuration	Highly agile, flexible configuration
Novelty/newness	Not new/matured	Very new/maturing
Security	Well-secured	Not yet well-secured
Users	Business professionals	Data scientists

processes that are tied to it. A data lake, on the other hand, lacks the structure of a data warehouse—which gives developers and data scientists the ability to easily configure and reconfigure their models, queries, and apps on-the-fly.

Novelty/newness. The technologies underlying data warehousing have been around for a long time. Most of the innovations have been accomplished in the last 20–30 years. Therefore, there is very little, if any, newness coming out of data warehousing (with the exclusion of the technologies to leverage and use Big Data within a data warehouse). On the other hand, data lakes are new and are going through a novelty/innovation phase to become a mainstream data storage technology.

Security. Because data warehouse technologies have been around for decades the ability to secure data in a data warehouse is much more mature than securing data in a data lake. It should be noted, however, that there is a significant effort being placed on security right now in the Big Data industry. It is not a question of if, but when, the security of the data lakes will meet the needs and wants of the analytics professionals and other end users.

Users. For a long time, the motto in the analytic world has been “Business intelligence and analytics for everyone!” We have built the data warehouse and invited “everyone” to come, but have they come? On average, only 20–25% of them have. Is it the same cry for the data lake? Will we build the data lake and invite everyone to come? Maybe in the future. For now, a data lake, at this point in its maturity, is best suited for the data scientists.

In summary, a data lake and a data warehouse are not the same. Also, the data lake is not Data Warehouse 2.0 (as suggested by some) or a replacement for the data warehouse. They both are needed and therefore optimized for different data mediums and different tasks/purposes. That is, they need to coexist in the analytics world (at least for a while—until data lakes mature to a level to accomplish what nowadays data warehouses are good for). The goal is to design and properly use each for what they were intended to do—use the best option for the job, which may turn out to be a hybrid of data warehouse and data lake storage mediums.

Sources: Inmon, B. (2021). Five Steps to a Successful Data Lakehouse, retrieved at databricks.com/wp-content/uploads/2021/10/The-Data-Lakehouse.pdf (accessed June 2022); Campbell, C. (2015). Top five differences between data lakes and data warehouses. www.blue-granite.com/blog/bid/402596/Top-Five-Differences-between-Data-Lakes-and-Data-Warehouses (accessed July 2022); Dull, T. (2021). Data lake vs data warehouse: Key differences. KDnuggets.com. <http://www.kdnuggets.com/2015/09/data-lake-vs-data-warehouse-key-differences.html> (accessed August 2016).

Infrastructure (architectural—hardware and software—enhancements):

- **Columnar (a new way to store and access data in the database).** A column-oriented database management system (also commonly called a *columnar database*) is a system that stores data tables as sections of columns of data rather than as rows of data (which is the way most RDBMS do it). That is, these columnar databases store data by columns instead of rows (all values of a single column are stored consecutively on disk memory). Such a structure gives a much finer grain of control to the RDBMS. It can access only the columns required for the query as opposed to being forced to access all columns of the row. It performs significantly better for queries that need a small percentage of the columns in the tables they are in, but performs significantly worse when you need most of the columns due to the overhead in attaching all of the columns together to form the result sets. Comparisons between row-oriented and column-oriented data layouts are typically concerned with the efficiency of hard disk access for a given workload (which happens to be one of the most time-consuming operations in a computer). Based on the task at hand, one may have significant advantages over the other. Column-oriented organizations are more efficient when (1) an aggregate needs to be computed over many rows but only for a notably smaller subset of all columns of data, because reading that smaller subset of data can be faster

than reading all data; and (2) new values of a column are supplied for all rows at once because that column data can be written efficiently and replace old column data without touching any other columns for the rows. Row-oriented organizations are more efficient when (1) many columns of a single row are required at the same time, and when row size is relatively small, as the entire row can be retrieved with a single-disk seek; and (2) writing a new row if all of the column data is supplied at the same time, as the entire row can be written with a single disk seek. In addition, because the data stored in a column is of uniform type, it lends itself better for compression. That is, significant storage size optimization is available in column-oriented data that is not available in row-oriented data. Such optimal compression of data reduces storage size, making it more economically justifiable to pursue in-memory or solid-state storage alternatives.

- **Real-time data warehousing.** **Real-time data warehousing (RDW)** implies that the refresh cycle of an existing data warehouse to update the data is more frequent (almost at the same time as the data becomes available at operational databases). These RDW systems can achieve near real-time updates of data, where the data latency typically is in the range from minutes to hours. As the latency gets smaller, the cost of data updates seems to be increasing exponentially. Future advancements in many technological fronts (ranging from automatic data acquisition to intelligent software agents) are needed to make RDW a reality with an affordable price tag.
- **Data warehouse appliances (all-in-one solutions to DW).** A data warehouse appliance consists of an integrated set of servers, storage, operating system(s), database management systems, and software specifically preinstalled and preoptimized for data warehousing. In practice, data warehouse appliances provide solutions for the mid-warehouse to Big Data warehouse market, offering low-cost performance on data volumes in the terabyte to petabyte range. To improve performance, most data warehouse appliance vendors use massively parallel processing architectures. Even though most database and data warehouse vendors provide appliances nowadays, many believe that Teradata was the first to provide a commercial data warehouse appliance product. What is most currently observed is the emergence of data warehouse bundles where vendors combine their hardware and database software as a data warehouse platform. From a benefits standpoint, data warehouse appliances have significantly low total cost of ownership, which includes initial purchase costs, ongoing maintenance costs, and the cost of changing capacity as the data grows. The resource cost for monitoring and tuning the data warehouse makes up a large part of the total cost of ownership, often as much as 80%. DW appliances reduce administration for day-to-day operations, setup, and integration. Because data warehouse appliances provide a single-vendor solution, they tend to better optimize the hardware and software within the appliance. Such a unified integration maximizes the chances of successful integration and testing of the DBMS, storage, and operating system by avoiding some of the compatibility issues that arise from multivendor solutions. A data warehouse appliance also provides a single point of contact for problem resolution and a much simpler upgrade path for both software and hardware.
- **Data management technologies and practices.** Some of the most pressing needs for a next-generation data warehouse platform involve technologies and practices that we generally don't think of as part of the platform. In particular, many users need to update the data management tools that process data for use through the data warehousing. The future holds strong growth for master data management (MDM). This relatively new, but extremely important, concept is gaining popularity for many reasons, including the following: (1) tighter integration with

operational systems demands MDM; (2) most data warehouses still lack MDM and data quality functions; and (3) regulatory and financial reports must be perfectly clean and accurate.

- ***In-database processing technology (putting the algorithms where the data is).*** In-database processing (also called *in-database analytics*) refers to the integration of the algorithmic extent of data analytics into data warehousing. By doing so, the data and the analytics that work off the data live within the same environment. Having the two in close proximity increases the efficiency of the computationally intensive analytics procedures. Today, many large database-driven decision support systems, such as those used for credit card fraud detection and investment bank risk management, use this technology because it provides significant performance improvements over traditional methods in a decision environment where time is of the essence. In-database processing is a complex endeavor compared to the traditional way of conducting analytics, where the data is moved out of the database (often in a flat-file format that consists of rows and columns) into a separate analytics environment (such as SAS Enterprise Modeler, Statistica Data Miner, or IBM SPSS Modeler) for processing. In-database processing makes more sense for high-throughput, real-time application environments, including fraud detection, credit scoring, risk management, transaction processing, pricing and margin analysis, usage-based micro-segmenting, behavioral ad targeting, and recommendation engines, such as those used by customer service organizations to determine next-best actions. In-database processing is performed and promoted as a feature by many of the major data warehousing vendors, including Teradata (integrating SAS analytics capabilities into the data warehouse appliances), IBM Netezza, EMC Greenplum, and Sybase, among others.
- ***In-memory storage technology (moving the data in the memory for faster processing).*** Conventional database systems, such as RDBMS, typically use physical hard drives to store data for an extended period of time. When a data-related process is requested by an application, the database management system loads the data (or parts of the data) into the main memory, processes it, and responds back to the application. Although data (or parts of the data) are cached temporarily in the main memory in a database management system, the primary storage location remains a magnetic hard disk. In contrast, an in-memory database system keeps the data permanently in the main memory. When a data-related process is requested by an application, the database management system directly accesses the data, which is already in the main memory, processes it, and responds back to the requesting application. This direct access to data in main memory makes the processing of data orders of magnitude faster than the traditional method. So the main benefit of in-memory technology (may be the only benefit of it) is the incredible speed at which it accesses the data. The disadvantages include the cost of paying for a very large main memory (even though it is getting cheaper, it still costs a great deal to have a large enough main memory to hold all of a company's data) and the need for sophisticated data recovery strategies (because main memory is volatile and can be wiped out accidentally).
- ***New database management systems.*** A data warehouse platform consists of several basic components, of which the most critical is the database management system (DBMS). This is only natural, given the fact that the DBMS is the component of the platform where the most work must be done to implement a data model and optimize it for query performance. Therefore, the DBMS is where many next-generation innovations are expected to happen.
- ***Advanced analytics.*** There are different analytic methods users can choose as they move beyond basic OLAP-based methods and into advanced analytics. Some

users choose advanced analytic methods based on data mining, predictive analytics, statistics, artificial intelligence, and so on. Still, the majority of users seem to be choosing SQL-based methods. Whether SQL-based or not, advanced analytics seem to be among the most important promises of next-generation data warehousing.

In this evolving big data era, the future of data warehousing seems to be full of promises, but at the same time, many uncertainties and significant challenges. Whatever the name may be, let it be data warehousing, data lakes, data lake houses, data reservoir, or any creative combination of these popular terms, the importance of collecting and analyzing data for timely and accurate decision making will not go away. In fact, as the world of business becomes more global and complex, the need for analytics and data warehousing tools will become more prominent. The fast-improving IT tools and techniques seem to be moving in the right direction to address the needs of the future decision support systems.

► SECTION 4.6 REVIEW QUESTIONS

1. What steps can an organization take to ensure the security and confidentiality of customer data in its data warehouse?
2. What skills should a DWA possess? Why?
3. What recent technologies may shape the future of data warehousing? Why?

4.7 BUSINESS REPORTING

Decision makers are in need of information to make accurate and timely decisions. Information is essentially the contextualization of data. In addition to statistical means that were explained in the previous section, information (descriptive analytics) can also be obtained using online analytics processing [OLTP] systems (see the simple taxonomy of descriptive analytics in Chapter 3, Figure 3.11). The information is usually provided to the decision makers in the form of a written report (digital or on paper), although it can also be provided orally. Simply put, a **report** is any communication artifact prepared with the specific intention of conveying information in a digestible form to whoever needs it, whenever and wherever they may need it. It is usually a document that contains information (usually driven from data) organized in a narrative, graphic, and/or tabular form, prepared periodically (recurring) or on an as-needed (ad hoc) basis, referring to specific time periods, events, occurrences, or subjects. Business reports can fulfill many different (but often related) functions. Here are a few of the most prevailing ones:

- To ensure that all departments are functioning properly
- To provide information
- To provide the results of an analysis
- To persuade others to act
- To create an organizational memory (as part of a knowledge management system)

Business reporting (also called OLAP or BI) is an essential part of the larger drive toward improved, evidence-based, optimal managerial decision making. The foundation of these **business reports** is various sources of data coming from both inside and outside the organization (online transaction processing [OLTP] systems). Creation of these reports involves ETL (extract, transform, and load) procedures in coordination with a data warehouse and then using one or more reporting tools (see Chapter 3 for a detailed description of these concepts).

Due to the rapid expansion of information technology coupled with the need for improved competitiveness in business, there has been an increase in the use of computing power to produce unified reports that join different views of the enterprise in one

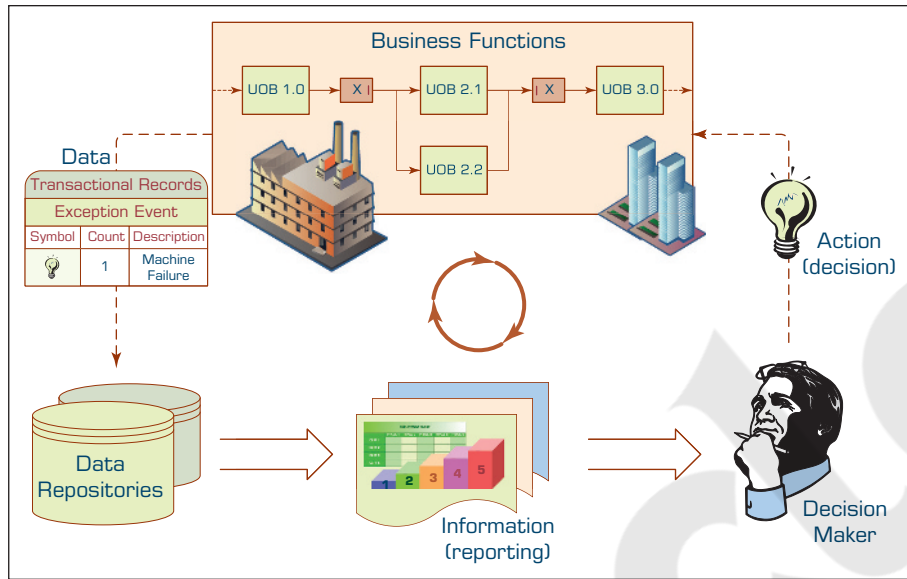


FIGURE 4.12 The Role of Information Reporting in Managerial Decision Making.

place. Usually, this reporting process involves querying structured data sources, most of which were created using different logical data models and data dictionaries, to produce a human-readable, easily digestible report. These types of business reports allow managers and coworkers to stay informed and involved, review options and alternatives, and make informed decisions. Figure 4.12 shows the continuous cycle of data acquisition → information generation → decision making → business process management. Perhaps the most critical task in this cyclical process is the reporting (i.e., information generation)—converting data from different sources into actionable information.

Key to any successful report are clarity, brevity, completeness, and correctness. The nature of the report and the level of importance of these success factors change significantly based on for whom the report is created. Most of the research in effective reporting is dedicated to internal reports that inform stakeholders and decision makers within the organization. There are also external reports between businesses and the government (e.g., for tax purposes or for regular filings to the Securities and Exchange Commission). Even though there are a wide variety of business reports, the ones that are often used for managerial purposes can be grouped into three major categories (Hill, 2016).

METRIC MANAGEMENT REPORTS In many organizations, business performance is managed through outcome-oriented metrics. For external groups, these are service-level agreements. For internal management, they are **key performance indicators (KPIs)**. Typically, there are enterprise-wide agreed targets to be tracked against over a period of time. They may be used as part of other management strategies such as Six Sigma or Total Quality Management.

DASHBOARD-TYPE REPORTS A popular idea in business reporting in recent years has been to present a range of different performance indicators on one page, like a dashboard in a car. Typically, dashboard vendors would provide a set of predefined reports with static elements and fixed structure, but also allow for customization of the dashboard widgets, views, and set targets for various metrics. It's common to have color-coded traffic lights defined for performance (red, orange, green) to draw management's attention to particular areas. A more detailed description of dashboards can be found in later part of this chapter.

BALANCED SCORECARD-TYPE REPORTS This is a method developed by Kaplan and Norton that attempts to present an integrated view of success in an organization. In addition to financial performance, balanced scorecard-type reports also include customer, business process, and learning and growth perspectives. More details on balanced scorecards are provided later in this chapter.

► SECTION 4.7 REVIEW QUESTIONS

1. What is a report? What are reports used for?
2. What is a business report? What are the main characteristics of a good business report?
3. Describe the cyclic process of management, and comment on the role of business reports.
4. List and describe the three major categories of business reports.
5. What are the main components of a business reporting system?

4.8 DATA VISUALIZATION

Data visualization (or more appropriately, information visualization) has been defined as “the use of visual representations to explore, make sense of, and communicate data” (Few, 2007). Although the name that is commonly used is *data visualization*, usually what is meant by this is information visualization. Because information is the aggregation, summarization, and contextualization of data (raw facts), what is portrayed in visualizations is the information and not the data. However, because the two terms *data visualization* and *information visualization* are used interchangeably and synonymously, in this chapter we will follow suit.

Data visualization is closely related to the fields of information graphics, information visualization, scientific visualization, and statistical graphics. Until recently, the major forms of data visualization available in both BI applications have included charts and graphs, as well as the other types of visual elements used to create scorecards and dashboards.

To better understand the current and future trends in the field of data visualization, it helps to begin with some historical context.

A Brief History of Data Visualization

Despite the fact that predecessors to data visualization date back to the second century AD, most developments have occurred in the last two and a half centuries, predominantly during the last 30 years (Few, 2007). Although visualization has not been widely recognized as a discipline until fairly recently, today’s most popular visual forms date back a few centuries. Geographical exploration, mathematics, and popularized history spurred the creation of early maps, graphs, and timelines as far back as the 1600s, but William Playfair is widely credited as the inventor of the modern chart, having created the first widely distributed line and bar charts in his *Commercial and Political Atlas of 1786* and what is generally considered to be one of the first time series portraying line charts in his *Statistical Breviary*, published in 1801 (see Figure 4.13).

Perhaps the most notable innovator of information graphics during this period was Charles Joseph Minard, who graphically portrayed the losses suffered by Napoleon’s army in the Russian campaign of 1812 (see Figure 4.14). Beginning at the Polish–Russian border, the thick band shows the size of the army at each position. The path of Napoleon’s retreat from Moscow in the bitterly cold winter is depicted by the dark lower band, which is tied to temperature and time scales. Popular visualization expert, author, and critic Edward Tufte

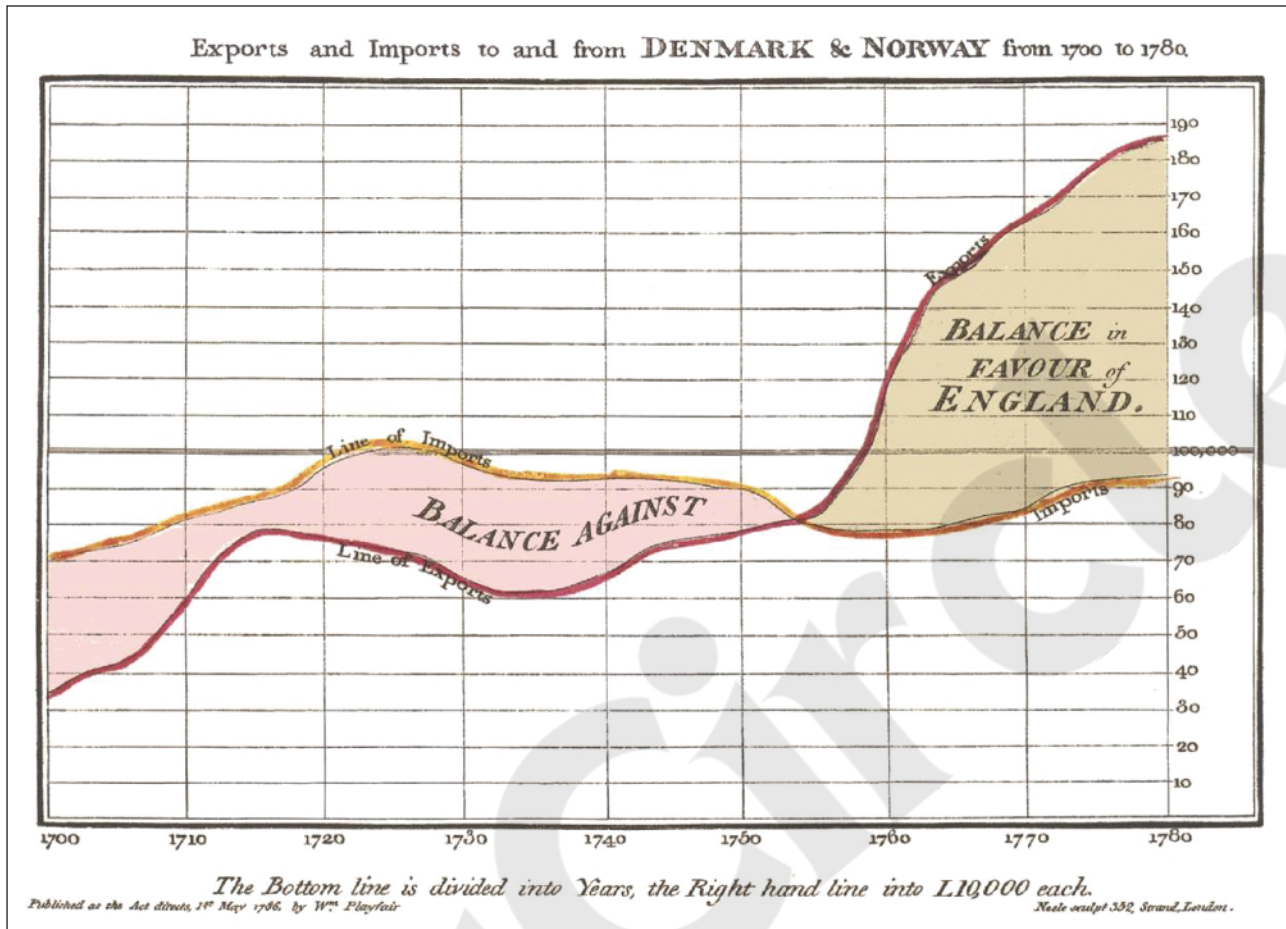


FIGURE 4.13 One of The First Time Series Line Chart (Created by William Playfair in 1801).

says that this “may well be the best statistical graphic ever drawn.” In this graphic Minard managed to simultaneously represent several data dimensions (the size of the army, direction of movement, geographic locations, outside temperature, etc.) in an artistic and informative manner. Many more excellent visualizations were created in the 1800s, and most of them are chronicled on Tufte’s Web site (edwardtufte.com) and his visualization books.

The 1900s saw the rise of a more formal, empirical attitude toward visualization, which tended to focus on aspects such as color, value scales, and labeling. In the mid-1900s, cartographer and theorist Jacques Bertin published his *Semiologie Graphique*, which some say serves as the theoretical foundation of modern information visualization. Although most of his patterns are either outdated by more recent research or completely inapplicable to digital media, many are still very relevant.

In the 2000s, the Internet emerged as a new medium for visualization and brought with it a whole lot of new tricks and capabilities. Not only has the worldwide, digital distribution of both data and visualization made them more accessible to a broader audience (raising visual literacy along the way), but it has also spurred the design of new forms that incorporate interaction, animation, and graphics-rendering technology unique to screen media, and real-time data feeds to create immersive environments for communicating and consuming data.

► SECTION 4.8 REVIEW QUESTIONS

1. What is data visualization? Why is it needed?
2. What are the historical roots of data visualization?
3. Carefully analyze Charles Joseph Minard's graphical portrayal of Napoleon's march. Identify and comment on all the information dimensions captured in this ancient diagram.
4. Who is Edward Tufte? Why do you think we should know about his work?
5. What do you think is the "next big thing" in data visualization?

4.9 DIFFERENT TYPES OF CHARTS AND GRAPHS

Often end users of business analytics systems are not sure what type of chart or graph to use for a specific purpose. Some charts or graphs are better at answering certain types of questions. Some look better than others. Some are simple; some are rather complex and crowded. What follows is a short description of the types of charts and/or graphs commonly found in most business analytics tools and what types of questions they are better at answering/analyzing. This material is compiled from several published articles and other literature (Abela, 2008; Hardin et al., 2012).

Basic Charts and Graphs

What follows are the basic charts and graphs that are commonly used for information visualization.

LINE CHART Line charts are perhaps the most frequently used graphical visuals for time series data. Line charts (or a line graphs) show the relationship between two variables; they are most often used to track changes or trends over time (having one of the variables set to time on the x -axis). Line charts sequentially connect individual data points to help infer changing trends over a period of time. Line charts are often used to show time-dependent changes in the values of some measure, such as changes on a specific stock price over a 5-year period or changes on the number of daily customer service calls over a month.

BAR CHART Bar charts are among the most basic visuals used for data representation. Bar charts are effective when you have nominal data or numerical data that splits nicely into different categories so you can quickly see comparative results and trends within your data. Bar charts are often used to compare data across multiple categories such as percent of advertising spending by departments or by product categories. Bar charts can be vertically or horizontally oriented. They can also be stacked on top of each other to show multiple dimensions in a single chart.

PIE CHART **Pie charts** are visually appealing, as the name implies, pie-looking charts. Because they are so visually attractive, they are often incorrectly used. Pie charts should only be used to illustrate relative proportions of a specific measure. For instance, they can be used to show the relative percentage of an advertising budget spent on different product lines, or they can show relative proportions of majors declared by college students in their sophomore year. If the number of categories to show is more than just a few (say more than four), one should seriously consider using a bar chart instead of a pie chart.

SCATTER PLOT **Scatter plots** are often used to explore the relationship between two or three variables (in 2-D or 2-D visuals). Because they are visual exploration tools, having more than three variables, translating them into more than three dimensions is not easily

achievable. Scatter plots are an effective way to explore the existence of trends, concentrations, and outliers. For instance, in a two-variable (two-axis) graph, a scatter plot can be used to illustrate the corelationship between age and weight of heart disease patients or it can illustrate the relationship between the number of customer care representatives and the number of open customer service claims. Often, a trend line is superimposed on a two-dimensional scatter plot to illustrate the nature of the relationship.

BUBBLE CHART **Bubble charts** are often enhanced versions of scatter plots. Bubble charts, though, are not a new visualization type; instead, they should be viewed as a technique to enrich data illustrated in scatter plots (or even geographic maps). By varying the size and/or color of the circles, one can add additional data dimensions, offering more enriched meaning about the data. For instance, a bubble chart can be used to show a competitive view of college-level class attendance by major and by time of the day, or it can be used to show profit margin by product type and by geographic region.

Specialized Charts and Graphs

The graphs and charts that we review in this section are either derived from the basic charts as special cases or they are relatively new and are specific to a problem type and/or an application area.

HISTOGRAM Graphically speaking, a **histogram** looks just like a bar chart. The difference between histograms and generic bar charts is the information that is portrayed. Histograms are used to show the frequency distribution of a variable or several variables. In a histogram, the *x*-axis is often used to show the categories or ranges, and the *y*-axis is used to show the measures/values/frequencies. Histograms show the distributional shape of the data. That way, one can visually examine if the data is normally or exponentially distributed. For instance, one can use a histogram to illustrate the exam performance of a class, where distribution of the grades as well as comparative analysis of individual results can be shown, or one can use a histogram to show age distribution of the customer base.

GANTT CHART Gantt charts are a special case of horizontal bar charts that are used to portray project timelines, project tasks/activity durations, and overlap among the tasks/activities. By showing start and end dates/times of tasks/activities and the overlapping relationships, Gantt charts provide an invaluable aid for management and control of projects. For instance, Gantt charts are often used to show project timelines, task overlaps, relative task completions (a partial bar illustrating the completion percentage inside a bar that shows the actual task duration), resources assigned to each task, milestones, and deliverables.

PERT CHART PERT charts (also called network diagrams) are developed primarily to simplify the planning and scheduling of large and complex projects. They show precedence relationships among the project activities/tasks. A PERT chart is composed of nodes (represented as circles or rectangles) and edges (represented with directed arrows). Based on the selected PERT chart convention, either nodes or the edges may be used to represent the project activities/tasks (activity-on-node versus activity-on-arrow representation schema).

GEOGRAPHIC MAP When the data set includes any kind of location data (e.g., physical addresses, postal codes, state names or abbreviations, country names, latitude/longitude, or some type of custom geographic encoding), it is better and more informative to see the data on a map. Maps usually are used in conjunction with other charts and graphs, as opposed to by themselves. For instance, one can use maps to show distribution of

customer service requests by product type (depicted in pie charts) by geographic locations. Often a large variety of information (e.g., age distribution, income distribution, education, economic growth, or population changes) can be portrayed in a geographic map to help decide where to open a new restaurant or a new service station. These types of systems are often called geographic information systems (GIS).

BULLET Bullet graphs are often used to show progress toward a goal. A bullet graph is essentially a variation of a bar chart. Often they are used in place of gauges, meters, and thermometers in a dashboard to more intuitively convey the meaning within a much smaller space. Bullet graphs compare a primary measure (e.g., year-to-date revenue) to one or more other measures (e.g., annual revenue target) and present this in the context of defined performance metrics (e.g., sales quotas). A bullet graph can intuitively illustrate how the primary measure is performing against overall goals (e.g., how close a sales representative is to achieving his/her annual quota).

HEAT MAP Heat maps are great visuals to illustrate the comparison of continuous values across two categories using color. The goal is to help the user quickly see where the intersection of the categories is strongest and weakest in terms of numerical values of the measure being analyzed. For instance, one can use heat maps to show segmentation analysis of target markets where the measure (color gradient would be the purchase amount) and the dimensions would be age and income distribution.

HIGHLIGHT TABLE Highlight tables are intended to take heat maps one step further. In addition to showing how data intersects by using color, highlight tables add a number on top to provide additional detail. That is, they are two-dimensional tables with cells populated with numerical values and gradients of colors. For instance, one can show sales representatives' performance by product type and by sales volume.

TREE MAP Tree maps display hierarchical (tree-structured) data as a set of nested rectangles. Each branch of the tree is given a rectangle, which is then tiled with smaller rectangles representing subbranches. A leaf node's rectangle has an area proportional to a specified dimension on the data. Often the leaf nodes are colored to show a separate dimension of the data. When the color and size dimensions are correlated in some way with the tree structure, one can often easily see patterns that would be difficult to spot in other ways, such as if a certain color is particularly relevant. A second advantage of tree maps is that, by construction, they make efficient use of space. As a result, they can legibly display thousands of items on the screen simultaneously.

Which Chart or Graph Should You Use?

Which chart or graph that we explained in the previous section is the best? The answer is rather easy: there is not one best chart or graph, because if there was we would not have these many chart and graph types. They all have somewhat different data representation "skills." Therefore, the right question should be, "Which chart or graph is the best for a given task?" The capabilities of the charts given in the previous section can help in selecting and using the right chart/graph for a specific task, but it still is not easy to sort out. Several different chart/graph types can be used for the same visualization task. One rule of thumb is to select and use the simplest one from the alternatives to make it easy for the intended audience to understand and digest.

Although there is not a widely accepted, all-encompassing chart selection algorithm or chart/graph taxonomy, Figure 4.15 presents a rather comprehensive and highly logical organization of chart/graph types in a taxonomy-like structure (the original version was

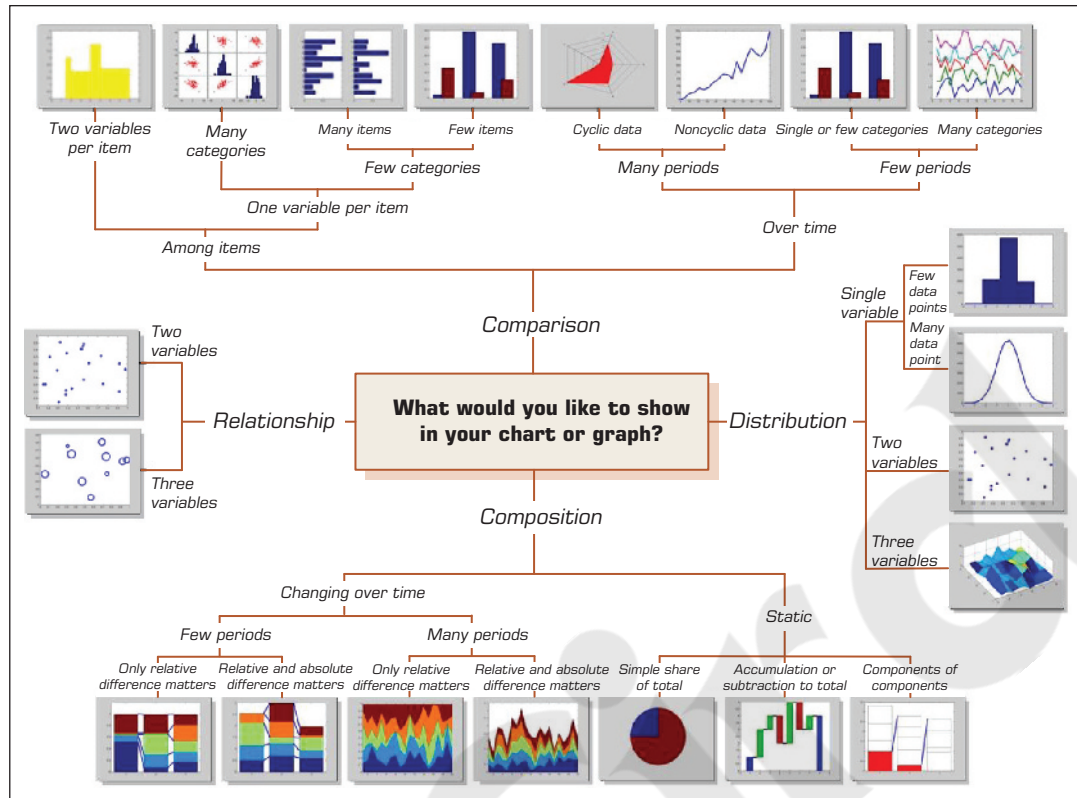


FIGURE 4.15 A Taxonomy of Charts and Graphs. Source: Adapted from Abela, A. (2008). *Advanced Presentations by Design: Creating Communication that Drives Action*. New York: Wiley.

published in Abela 2008). The taxonomic structure is organized around the questions of “What would you like to show in your chart or graph?” That is, what the purpose of the chart or graph will be. At that level, the taxonomy divides the purpose into four different types—relationship, comparison, distribution, and composition—and further divides the branches into subcategories based on the number of variables involved and time dependency of the visualization.

Even though these charts and graphs cover a major part of what is commonly used in information visualization, they by no means cover it all. Nowadays, one can find many other specialized graphs and charts that serve a specific purpose. Furthermore, the current trend is to combine/hybridize and animate these charts for better-looking and more intuitive visualization of today’s complex and volatile data sources. For instance, the interactive, animated, bubble charts available at the Gapminder Web site (gapminder.org) provide an intriguing way of exploring world health, wealth, and population data from a multidimensional perspective. Figure 4.16 depicts the sorts of displays available at the site. In this graph, population size, life expectancy, and per capita income at the continent level are shown; also given is a time-varying animation that shows how these variables change over time.

SECTION 4.9 REVIEW QUESTIONS

1. Why do you think there are many different types of charts and graphs?
2. What are the main differences among line, bar, and pie charts? When should you use one over the others?

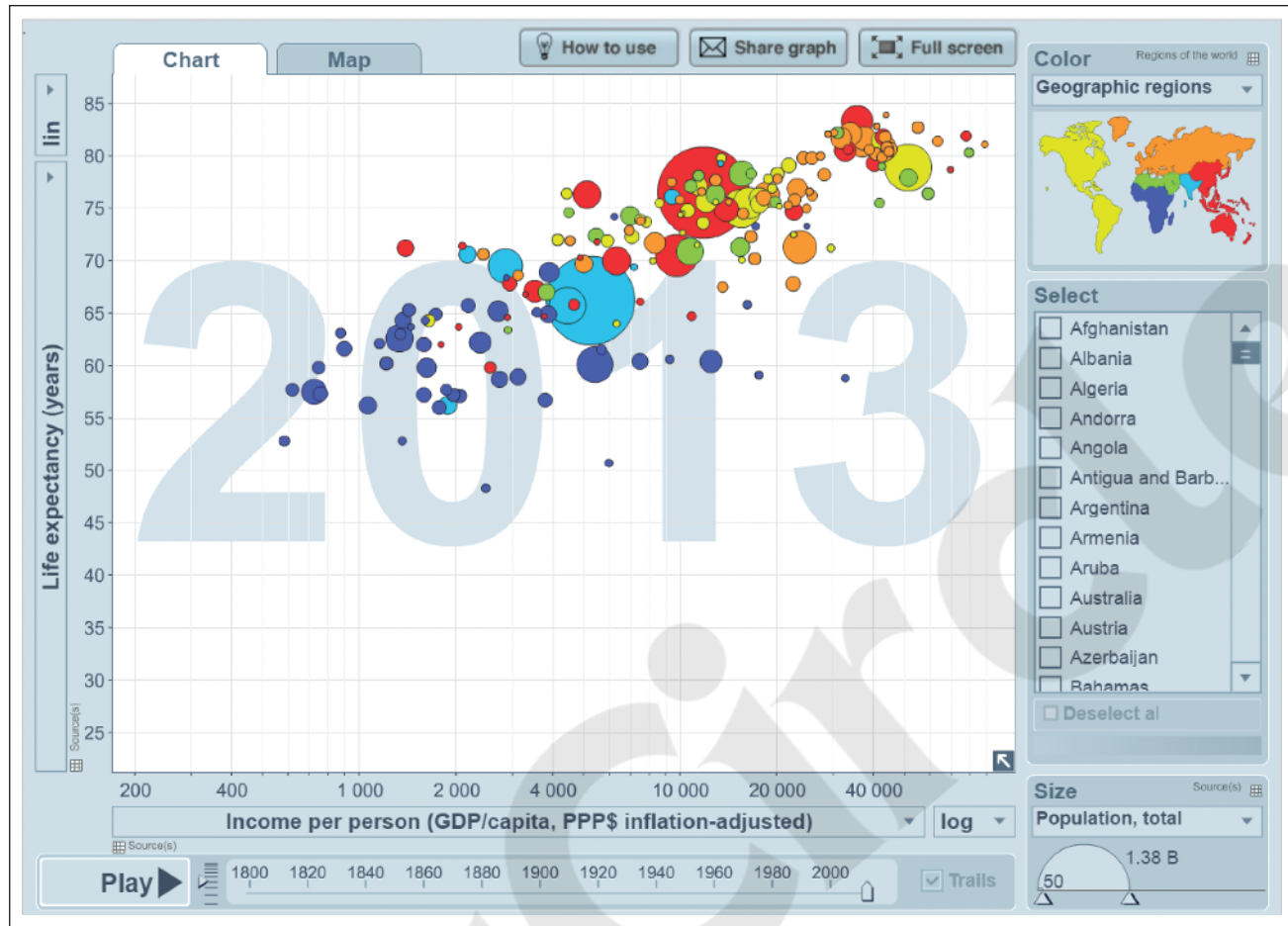


FIGURE 4.16 A Gapminder Chart that Shows the Wealth and Health of Nations. Source: gapminder.org.

3. Why would you use a geographic map? What other types of charts can be combined with a geographic map?
4. Find and explain the role of two types of charts that are not covered in this section.

4.10 THE EMERGENCE OF VISUAL ANALYTICS

As Seth Grimes (2009a,b) has noted, there is a “growing palate” of data visualization techniques and tools that enable the users of business analytics and BI systems to better “communicate relationships, add historical context, uncover hidden correlations, and tell persuasive stories that clarify and call to action.” The latest Magic Quadrant on Business Intelligence and Analytics Platforms released by Gartner in February 2016 further emphasizes the importance of data visualization in BI and analytics. As the chart shows, all the solution providers in the *Leaders* and *Visionary* quadrants are either relatively recently founded information visualization companies (e.g., Tableau Software, QlikTech) or well-established large analytics companies (e.g., Microsoft, SAS, IBM, SAP, MicroStrategy, Alteryx) that are increasingly focusing their efforts on information visualization and visual analytics. More details on Gartner’s latest Magic Quadrant are given in Technology Insights 4.3.

TECHNOLOGY INSIGHTS 4.3 Gartner Magic Quadrant for Business Intelligence and Analytics Platforms

Gartner, Inc., the creator and publisher of Magic Quadrants, is the leading information technology research and advisory company publically traded in the United States with over \$4.7 billion annual revenues in 2021. Founded in 1979, Gartner has 16,600 associates, a significant portion of which are research analysts and consultants, and numerous clients in more than 100 countries.

Magic Quadrant is a research method designed and implemented by Gartner to monitor and evaluate the progress and positions of companies in a specific, technology-based market. By applying a graphical treatment and a uniform set of evaluation criteria, Magic Quadrant helps users to understand how technology providers are positioned within a market.

Gartner changed the name of this Magic Quadrant from “Business Intelligence Platforms” to “Business Intelligence and Analytics Platforms” to emphasize the growing importance of analytics capabilities to the information systems that organizations are now building. Gartner defines the BI and analytics platform market as a software platform that delivers 15 capabilities across three categories: integration, information delivery, and analysis. These capabilities enable organizations to build precise systems of classification and measurement to support decision making and improve performance.

Figure 4.17 illustrates the latest Magic Quadrant for Business Intelligence and Analytics Platforms. Magic Quadrant places providers in four groups (niche players, challengers,

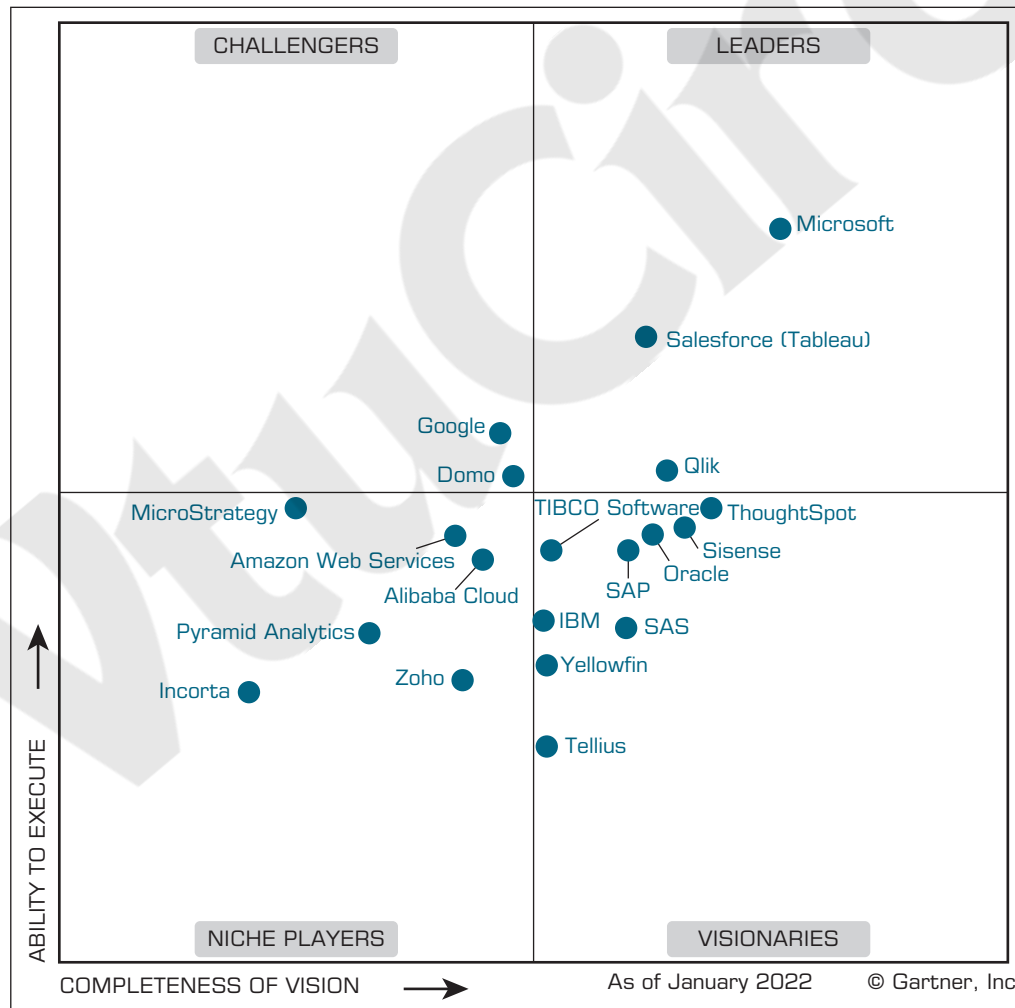


FIGURE 4.17 Magic Quadrant for Business Intelligence and Analytics Platforms. Source: gartner.com.

visionaries, and leaders) along two dimensions: completeness of vision (x -axis) and ability to execute (y -axis). As the quadrant clearly shows, most of the well-known data visualization, visual analytics, BI/BA providers are positioned in the “leaders” category while many of the lesser known, relatively new, emerging providers are positioned in the “niche players” category.

The BI and analytics platform market’s multiyear shift from IT-led enterprise reporting to business-led self-service analytics seems to have passed the tipping point. Most new buying is of modern, business-user-centric visual analytics platforms forcing a new market perspective, significantly reordering the vendor landscape. Most of the activity in the BI and analytics platform market is from organizations that are trying to mature their visualization capabilities and to move from descriptive to predictive and prescriptive analytics echelons. The vendors in the market have overwhelmingly concentrated on meeting this user demand. If there were a single market theme in 2022, it would be that data discovery/visualization became a mainstream architecture. While data discovery/visualization vendors such as Microsoft (with PowerBI), Salesforce (with Tableau), and Qlik are solidifying their position in the *Leaders* quadrant, others (both emerging and large, well-established tool/solution providers) are trying to move out of *Visionaries* into the *Leaders* quadrant.

This emphasis on data discovery/visualization from most of the leaders and visionaries in the market—which are now promoting tools with business-user-friendly data integration, coupled with embedded storage and computing layers and unfettered drilling—continues to accelerate the trend toward decentralization and user empowerment of BI and analytics and greatly enables organizations’ ability to perform diagnostic analytics.

Source: Gartner Magic Quadrant, released in February 2022, gartner.com (accessed June 2022).

In BI and analytics, the key challenges for visualization have revolved around the intuitive representation of large, complex data sets with multiple dimensions and measures. For the most part, the typical charts, graphs, and other visual elements used in these applications usually involve two dimensions, sometimes three, and fairly small subsets of data sets. In contrast, the data in these systems reside in a data warehouse. At a minimum, these warehouses involve a range of dimensions (e.g., product, location, organizational structure, time), a range of measures, and millions of cells of data. In an effort to address these challenges, a number of researchers have developed a variety of new visualization techniques.

Visual Analytics

Visual analytics is a recently coined term that is often used loosely to mean nothing more than information visualization. What is meant by **visual analytics** is the combination of visualization and predictive analytics. Whereas information visualization is aimed at answering, “What happened?” and “What is happening?” and is closely associated with BI (routine reports, scorecards, and dashboards), visual analytics is aimed at answering, “Why is it happening?” “What is more likely to happen?” and is usually associated with business analytics (forecasting, segmentation, correlation analysis). Many of the information visualization vendors are adding the capabilities to call themselves visual analytics solution providers. One of the top, long time analytics solution providers, SAS Institute, is approaching it from another direction. They are embedding their analytics capabilities into a high-performance data visualization environment that they call visual analytics.

Visual or not visual, automated or manual, online or paper based, business reporting is not much different than telling a story. Technology Insights 4.4 provides a different, unorthodox viewpoint to better business reporting.

TECHNOLOGY INSIGHTS 4.4 Telling Great Stories with Data and Visualization

Everyone who has data to analyze has stories to tell, whether it's diagnosing the reasons for manufacturing defects, selling a new idea in a way that captures the imagination of your target audience, or informing colleagues about a particular customer service improvement program. And when it's telling the story behind a big strategic choice so that you and your senior management team can make a solid decision, providing a fact-based story can be especially challenging. In all cases, it's a big job. You want to be interesting and memorable; you know you need to keep it simple for your busy executives and colleagues. Yet you also know you have to be factual, detail oriented, and data driven, especially in today's metric-centric world.

It's tempting to present just the data and facts, but when colleagues and senior management are overwhelmed by data and facts without context, you lose. We have all experienced presentations with large slide decks, only to find that the audience is so overwhelmed with data that they don't know what to think, or they are so completely tuned out that they take away only a fraction of the key points.

Start engaging your executive team and explaining your strategies and results more powerfully by approaching your assignment as a story. You will need the "what" of your story (the facts and data) but you also need the "Who?" "How?" "Why?" and the often-missed "So what?" It's these story elements that will make your data relevant and tangible for your audience. Creating a good story can aid you and senior management in focusing on what is important.

Why Story?

Stories bring life to data and facts. They can help you make sense and order out of a disparate collection of facts. They make it easier to remember key points and can paint a vivid picture of what the future can look like. Stories also create interactivity—people put themselves into stories and can relate to the situation.

Cultures have long used **storytelling** to pass on knowledge and content. In some cultures, storytelling is critical to their identity. Among the many examples from disparate parts of the world, this one is from New Zealand, where some of the Maori people tattoo their faces with *mokus*. A *moku* is a facial tattoo containing a story about ancestors—the family tribe. A man may have a tattoo design on his face that shows features of a hammerhead to highlight unique qualities about his lineage. The design he chooses signifies what is part of his "true self" and his ancestral home.

Likewise, when we are trying to understand a story, the storyteller navigates to finding the "true north." If senior management is looking to discuss how they will respond to a competitive change, a good story can make sense and order out of a lot of noise. For example, you may have facts and data from two studies, one including results from an advertising study and one from a product satisfaction study. Developing a story for what you measured across both studies can help people see the whole where there were disparate parts. For rallying your distributors around a new product, you can employ a story to give vision to what the future can look like. Most important, storytelling is interactive—typically the presenter uses words and pictures that audience members can put themselves into. As a result, they become more engaged and better understand the information.

So What Is a Good Story?

Most people can easily rattle off their favorite film or book. Or they remember a funny story that a colleague recently shared. Why do people remember these stories? Because they contain certain characteristics. First, a good story has great characters. In some cases, the reader or viewer has a vicarious experience where they become involved with the character. The character then has to be faced with a challenge that is difficult but believable. There must be hurdles that the character overcomes. And finally, the outcome or prognosis is clear by the end of the story. The situation may not be resolved—but the story has a clear endpoint.

Think of Your Analysis as a Story—Use a Story Structure

When crafting a data-rich story, the first objective is to find the story. Who are the characters? What is the drama or challenge? What hurdles have to be overcome? And at the end of your story, what do you want your audience to do as a result?

Once you know the core story, craft your other story elements: define your characters, understand the challenge, identify the hurdles, and crystallize the outcome or decision question. Make sure you are clear with what you want people to do as a result. This will shape how your audience will recall your story. With the story elements in place, write out the storyboard, which represents the structure and form of your story. Although it's tempting to skip this step, it is better first to understand the story you are telling and then to focus on the presentation structure and form. Once the storyboard is in place, the other elements will fall into place. The storyboard will help you to think about the best analogies or metaphors, to clearly set up challenge or opportunity, and to finally see the flow and transitions needed. The storyboard also helps you focus on key visuals (graphs, charts, and graphics) that you need your executives to recall. Figure 2.24 shows a storyline for the impact of small loans in a worldwide view within the Tableau visual analytics environment.

In summary, don't be afraid to use data to tell great stories. Being factual, detail oriented, and data driven is critical in today's metric-centric world, but it does not have to mean being boring and lengthy. In fact, by finding the real stories in your data and following the best practices, you can get people to focus on your message—and thus on what's important. Here are those best practices:

- Think of your analysis as a story—use a story structure.
- Be authentic—your story will flow.
- Be visual—think of yourself as a film editor.
- Make it easy for your audience and you.
- Invite and direct discussion.

Sources: Fink, E., & Moore, S. J. (2022). 5 Best Practices for Telling Great Data Stories—And Why It Will Make You a Better Analyst. White paper by Tableau Software, Inc., <https://www.tableau.com/whitepapers/telling-data-stories> (accessed June 2022); Knaflitz, C. N. (2015). *Storytelling with Data: A Data Visualization Guide for Business Professionals*. John Wiley & Sons.

High-Powered Visual Analytics Environments

Due to the increasing demand for visual analytics coupled with fast-growing data volumes, there is an exponential movement toward investing in highly efficient visualization systems. With their latest move into visual analytics, the statistical software giant SAS Institute is now among those who are leading this wave. Their new product, SAS Visual Analytics, is a very **high-performance computing**, in-memory solution for exploring massive amounts of data in a very short time (almost instantaneously). It empowers users to spot patterns, identify opportunities for further analysis, and convey visual results via Web reports or a mobile platform such as tablets and smartphones. Figure 4.18 shows the high-level architecture of the SAS Visual Analytics platform. On one end of the architecture, there is a universal data builder and administrator capabilities, leading into explorer, report designer, and mobile BI modules, collectively providing an end-to-end visual analytics solution.

Some of the key benefits proposed by SAS analytics are the following:

- Empowers all users with data exploration techniques and approachable analytics to drive improved decision making. SAS Visual Analytics enables different types of users to conduct fast, thorough explorations on all available data. Sampling to reduce the data is not required and not preferred.
- Easy-to-use, interactive Web interfaces broaden the audience for analytics, enabling everyone to glean new insights. Users can look at more options, make more precise decisions, and drive success even faster than before.
- Answer complex questions faster, enhancing the contributions from your analytic talent. SAS Visual Analytics augments the data discovery and exploration process by

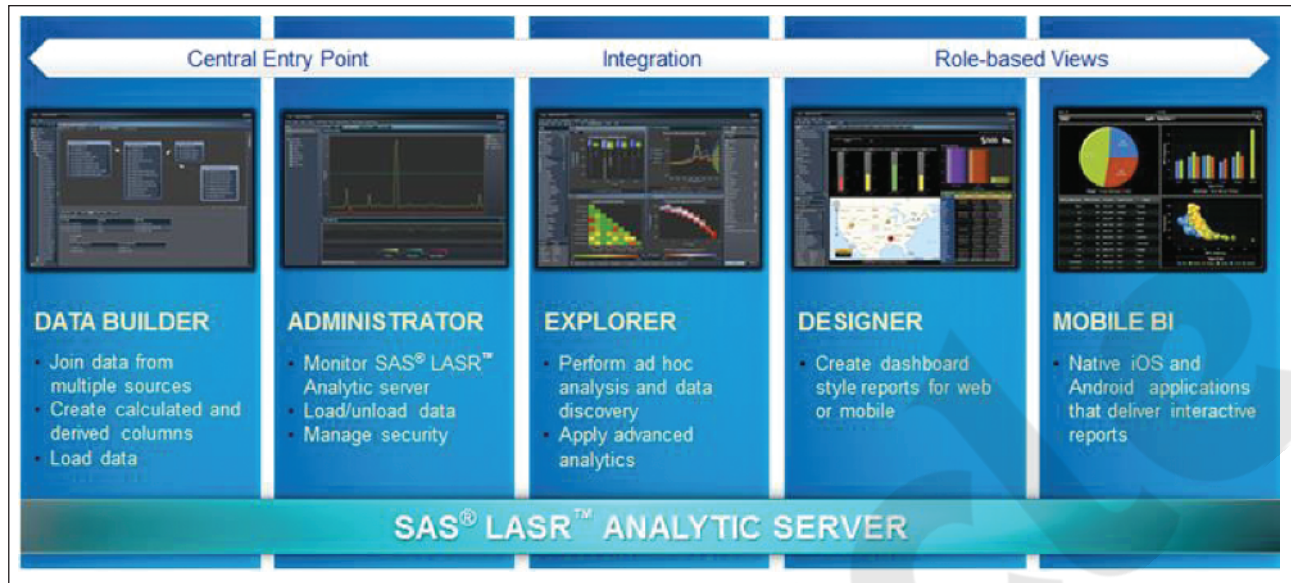


FIGURE 4.18 An Overview of SAS Visual Analytics Architecture. Source: SAS.com.

providing extremely fast results to enable better, more focused analysis. Analytically savvy users can identify areas of opportunity or concern from vast amounts of data so further investigation can take place quickly.

- Improves information sharing and collaboration. Large numbers of users, including those with limited analytical skills, can quickly view and interact with reports and charts via the Web, Adobe PDF files, and iPad mobile devices, while IT maintains control of the underlying data and security. SAS Visual Analytics provides the right information to the right person at the right time to improve productivity and organizational knowledge.
- Liberates IT by giving users a new way to access the information they need. Frees IT from the constant barrage of demands from users who need access to different amounts of data, different data views, ad hoc reports, and one-off requests for information. SAS Visual Analytics enables IT to easily load and prepare data for multiple users. Once data is loaded and available, users can dynamically explore data, create reports, and share information on their own.
- Provides room to grow at a self-determined pace. SAS Visual Analytics provides the option of using commodity hardware or database appliances from EMC Greenplum and Teradata. It is designed from the ground up for performance optimization and scalability to meet the needs of any size organization.

Figure 4.19 shows a screenshot of an SAS Analytics platform where time series forecasting and confidence interval around the forecast are depicted.

► SECTION 4.10 REVIEW QUESTIONS

1. What are the main reasons for the recent emergence of visual analytics?
2. Look at Gartner's Magic Quadrant for Business Intelligence and Analytics Platforms. What do you see? Discuss and justify your observations.
3. What is the difference between information visualization and visual analytics?
4. Why should storytelling be a part of your reporting and data visualization?
5. What is a high-powered visual analytics environment? Why do we need it?

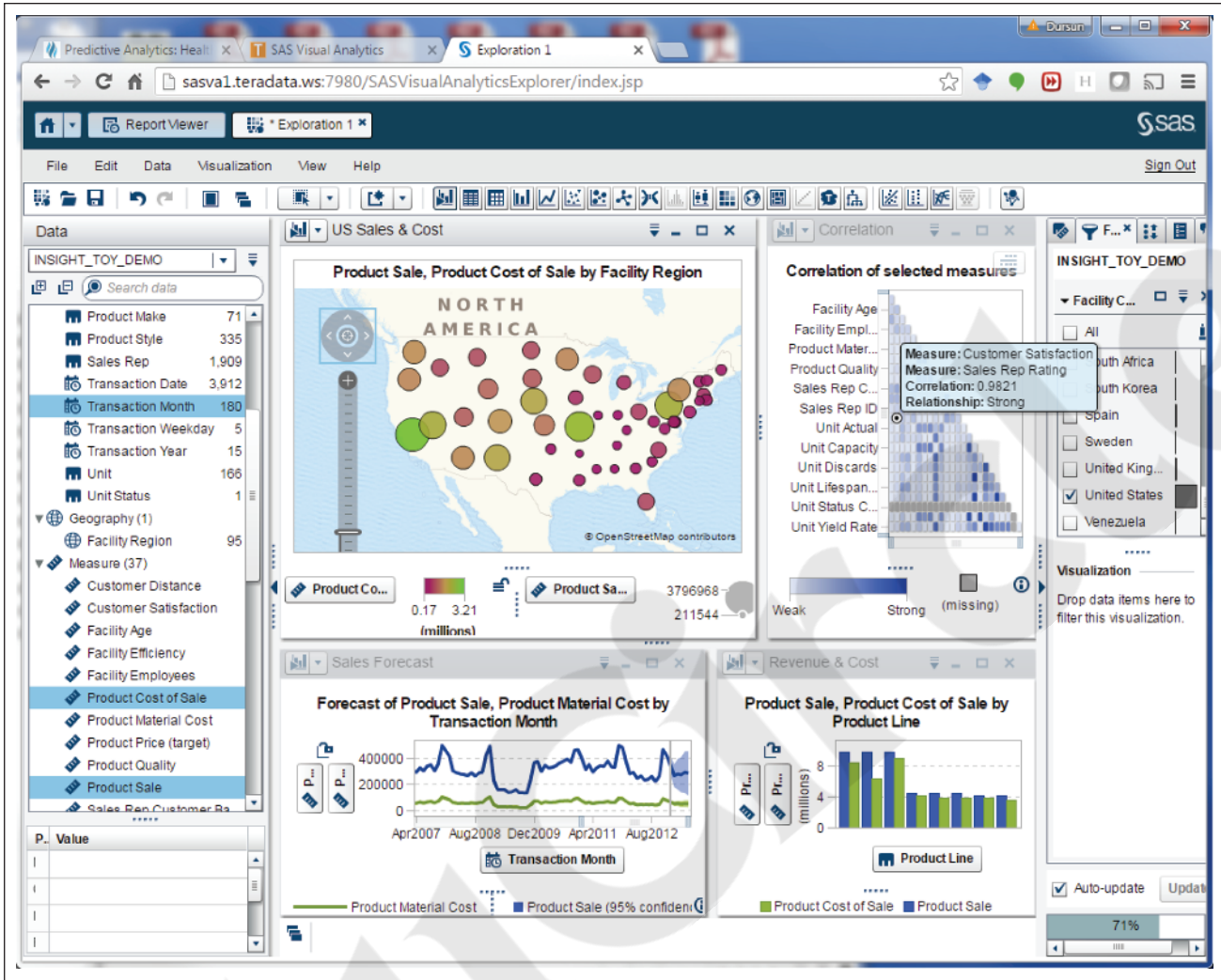


FIGURE 4.19 A Screenshot from SAS Visual Analytics. Source: SAS.com.

4.11 INFORMATION DASHBOARDS

Information dashboards are common components of most, if not all, BI or business analytics platforms, business performance management systems, and performance measurement software suites. **Dashboards** provide visual displays of important information that is consolidated and arranged on a single screen so that information can be digested at a single glance and easily drilled in and further explored. A typical dashboard is shown in Figure 4.20. This particular executive dashboard displays a variety of KPIs for a hypothetical software company called Sonatica (selling audio tools). This executive dashboard shows a high-level view of the different functional groups surrounding the products, starting from a general overview to the marketing efforts, sales, finance, and support departments. All of this is intended to give executive decision makers a quick and accurate idea of what is going on within the organization. On the left side of the dashboard, we can see (in a time series fashion) the quarterly changes in revenues, expenses, and margins, as well as the comparison of those figures to previous years'

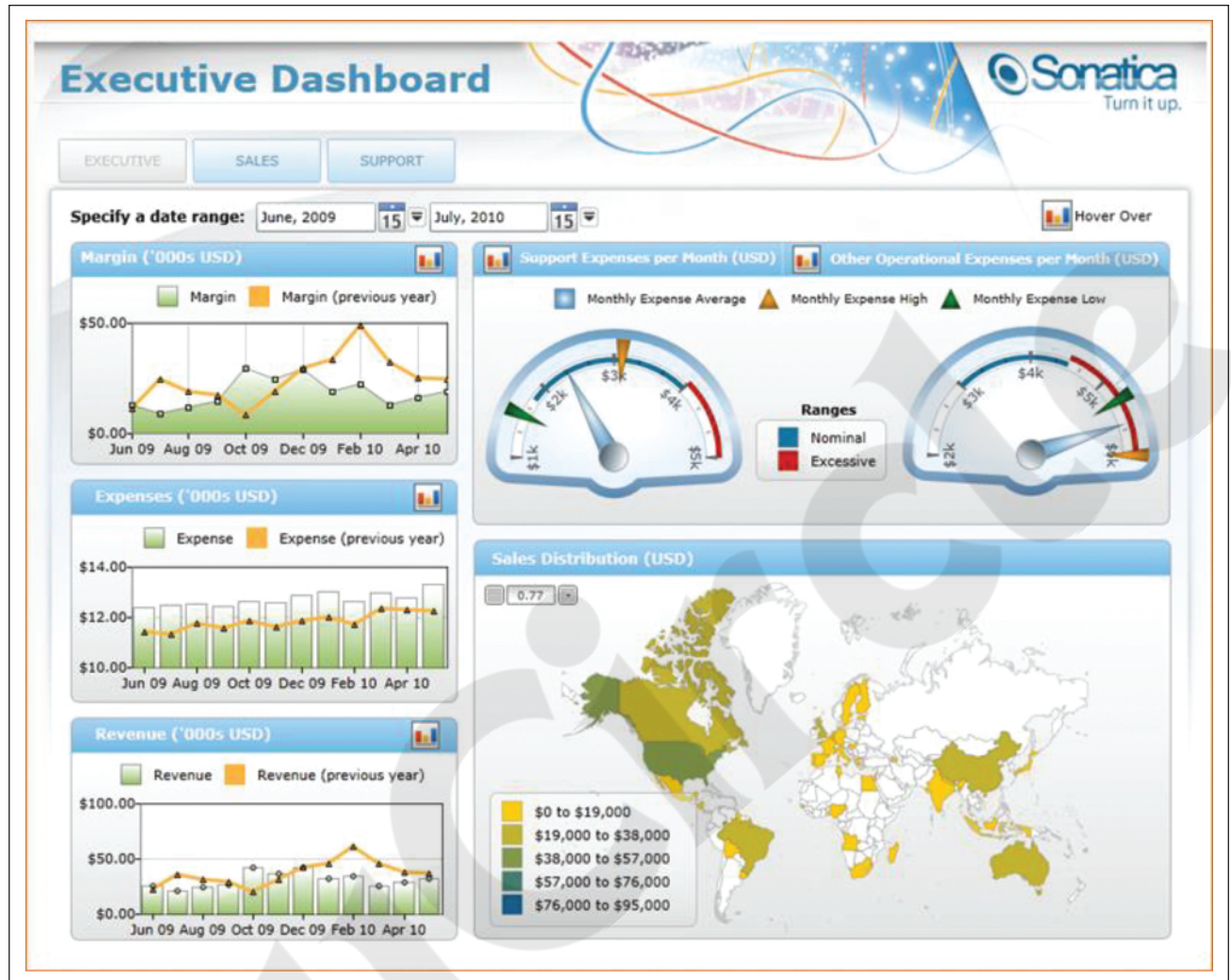


FIGURE 4.20 A Sample Executive Dashboard. Source: dundas.com.

monthly numbers. On the upper-right side we see two dials with color-coded regions showing the amount of monthly expenses for support services (dial on the left) and the amount of other expenses (dial on the right). As the color coding indicates, although the monthly support expenses are well within the normal ranges, the other expenses are in the red region, indicating excessive values. The geographic map on the bottom right shows the distribution of sales at the country level throughout the world. Behind these graphical icons, there are a variety of mathematical functions aggregating numerous data points to their highest level of meaningful figures. By clicking on these graphical icons, the consumer of this information can drill down to more granular levels of information and data.

Nowadays, a significant portion of the digitized data for business reports and dashboards come from the Internet, especially from social media. In Analytics in Action 4.3, you will find the summary of a successful implementation of data blending, efficient reports, and better informed decisioning.

ANALYTICS IN ACTION 4.3 Increasing the Efficiency of Social Media Campaign Reporting to Get to Insights Quicker

Motivation

Sport England is an arms-length body of government responsible for growing and developing grassroots sport and getting more people active across England. They're working to keep movement, sport, and physical activity central to the lives of everyone.

Sport England run cross-platform media campaigns to encourage participation in sport and physical activity. Reporting, however, is always a manual task. Sport England wanted to increase the efficiency of this process to not only generate insights more quickly, but also reduce unnecessary manual work.

Previously, the process required logging in to each of the social media platforms, collecting statistics from accounts, transforming the data so it was comparable, before visualizing it in Excel and producing a report in PowerPoint. This was a labor-intensive process, which could only be done by one person and created a significant lag in reporting. The delay in reporting meant that while a view of overall success was available to senior management, information was not available for operational decisions where a nimble response to events might be required. Off the shelf solutions were considered but could not provide the targeted approach or custom visuals required by Sport England for their "This Girl Can" campaign.

Implementation (Blending Technologies for an Automated, Bespoke, and Branded Reporting Solution)

Atos (an international IT services company and a trusted KNIME partner) created workflows in KNIME Analytics Platform to collect, transform, join, and write data from social media platforms to an SQL database on Azure, where it was visualized in Power BI. KNIME Server was used to schedule data collection and an analytical application was made available via the KNIME WebPortal to allow a supplementary data file to be uploaded in a user-friendly way by their admin team.

Atos used functionality already built into KNIME to connect to Twitter, Python nodes to connect to, navigate, and retrieve data from Facebook and Instagram graph databases, as well as Web-scraping for data from Medium. Python nodes were also used to query Google and YouTube APIs. Data was transformed and joined in KNIME to provide a comparable snapshot of activity over the previous 24 hours when the collected data ranged from cumulative counts to six-week windows.

The workflows were scheduled to run shortly after midnight on KNIME Server, which was running in a virtual machine on Sport England's Azure platform. Data was saved to an SQL database where it was available to Power BI (using the KNIME PowerBI Integration) for visualization.

Initially Azure Cognitive Services were to perform sentiment analysis on tweets in Power BI, but this functionality was folded back into KNIME. Therefore, a tweet was only ever passed to Cognitive Services once, and the score was stored in the database for cost and efficiency savings.

Atos created custom metrics for rating features such as a user's influence, which could be explained, understood, and adapted as necessary. These were then visualized to Sport England's specification in adherence with their internal style guides and branding to make it their tool. This highlighted that the data that was important to them and the success of their campaign.

Results (Automatic Data Processing and Increased Reporting Frequency)

The frequency of reporting was increased from monthly to daily. Instead of relying on a single person to generate the report, which would take half a day, the latest data was collected and processed automatically and made available via a Power BI report at the beginning of the day. The Power BI report was more useful than the PowerPoint report because it allowed users to apply multiple filters to find the data they needed—instead of relying on the analyst to anticipate requirements and generate insight.

Why KNIME?

KNIME Analytics Platform makes it easy to seamlessly blend core functionality for data manipulation with highly complex and bespoke operations for data collection and cleaning using R and Python. The visual workflow builder enables workflow creators to explain the process to the media team and others outside data specialist roles who don't necessarily have the technical data science knowledge, drawing parallels to how data was treated in the manual process. This means the solution is less of a "black box," but rather something that those who use the data can have confidence in because they understand how the parts fit together.

KNIME Server makes scheduling data collection a simple task and provides feedback that the workflows are executed without errors. Additional benefits include the ability to work remotely in the cloud, and the ability for Sport England to easily manage the data governance to sandbox this project from their other data.

Source: KNIME (2023). "Combining the Power of KNIME and PowerBI for Automated Sentiment Analysis" can be found at <https://www.knime.com/solutions/success-story/automated-sentiment-analysis> (accessed May 2023).

Dashboard Design

Dashboards are not a new concept. Their roots can be traced at least to the executive information system of the 1980s. Today, dashboards are ubiquitous. For example, a few years back, Forrester Research estimated that over 40% of the largest 2,000 companies in the world used the technology (Ante & McGregor, 2006). Since then, one can safely assume that this number has gone up quite significantly. In fact, nowadays it would be rather unusual to see a large company using a BI system that does not employ some sort of performance dashboards. The Dashboard Spy Web site (dashboardspy.com/about) provides further evidence of their ubiquity. The site contains descriptions and screenshots of thousands of BI dashboards, scorecards, and BI interfaces used by businesses of all sizes and industries, nonprofits, and government agencies.

According to Eckerson (2006), a well-known expert on BI in general and dashboards in particular, the most distinctive feature of a dashboard is its three layers of information:

- **Monitoring:** Graphical, abstracted data to monitor key performance metrics.
- **Analysis:** Summarized dimensional data to analyze the root cause of problems.
- **Management:** Detailed operational data that identify what actions to take to resolve a problem.

Because of these layers, dashboards pack a lot of information into a single screen. According to Few (2005), "The fundamental challenge of dashboard design is to display all the required information on a single screen, clearly and without distraction, in a manner that can be assimilated quickly." To speed assimilation of the numbers, the numbers need to be placed in context. This can be done by comparing the numbers of interest to other baseline or target numbers, by indicating whether the numbers are good or bad, by denoting whether a trend is better or worse, and by using specialized display widgets or components to set the comparative and evaluative context. Some of the common comparisons that are typically made in BI systems include comparisons against past values, forecasted values, targeted values, benchmark or average values, multiple instances of the same measure, and the values of other measures (e.g., revenues versus costs).

Even with comparative measures, it is important to specifically point out whether a particular number is good or bad and whether it is trending in the right direction. Without

these types of evaluative designations, it can be time consuming to determine the status of a particular number or result. Typically, either specialized visual objects (e.g., traffic lights, dials, and gauges) or visual attributes (e.g., color coding) are used to set the evaluative context.

What to Look for in a Dashboard

Although performance dashboards and other information visualization frameworks differ, they all do share some common design characteristics. First, they all fit within the larger BI and/or performance measurement system. This means that their underlying architecture is the BI or performance management architecture of the larger system. Second, all well-designed dashboard and other information visualizations possess the following characteristics (Novell, 2009):

- They use visual components (e.g., charts, performance bars, sparklines, gauges, meters, stoplights) to highlight, at a glance, the data and exceptions that require action.
- They are transparent to the user, meaning that they require minimal training and are extremely easy to use.
- They combine data from a variety of systems into a single, summarized, unified view of the business.
- They enable drill-down or drill-through to underlying data sources or reports, providing more detail about the underlying comparative and evaluative context.
- They present a dynamic, real-world view with timely data refreshes, enabling the end user to stay up to date with any recent changes in the business.
- They require little, if any, customized coding to implement, deploy, and maintain.

Best Practices in Dashboard Design

The real estate saying “location, location, location” makes it obvious that the most important attribute for a piece of real estate property is where it is located. For dashboards, it is “data, data, data.” An often overlooked aspect, data is one of the most important things to consider in designing dashboards. Even if a dashboard’s appearance looks professional, is aesthetically pleasing, and includes graphs and tables created according to accepted visual design standards, it is also important to ask about the data: Is it reliable? Is it timely? Is any data missing? Is it consistent across all dashboards? Here are some of the experience-driven best practices in dashboard design (Radha, 2008).

Benchmark Key Performance Indicators with Industry Standards

Many customers, at some point in time, want to know if the metrics they are measuring are the right metrics to monitor. Sometimes customers have found that the metrics they are tracking are not the right ones to track. Doing a gap assessment with industry benchmarks aligns you with industry best practices.

Wrap the Dashboard Metrics with Contextual Metadata

Often when a report or a visual dashboard/scorecard is presented to business users, questions remain unanswered. The following are some examples:

- Where did you source this data from?
- While loading the data warehouse, what percentage of the data got rejected/encountered data quality problems?

- Is the dashboard presenting “fresh” information or “stale” information?
- When was the data warehouse last refreshed?
- When is it going to be refreshed next?
- Were any high-value transactions that would skew the overall trends rejected as a part of the loading process?

Validate the Dashboard Design by a Usability Specialist

In most dashboard environments, the dashboard is designed by a tool specialist without giving consideration to usability principles. Even though it's a well-engineered data warehouse that can perform well, many business users do not use the dashboard, as it is perceived as not being user friendly, leading to poor adoption of the infrastructure and change management issues. Up-front validation of the dashboard design by a usability specialist can mitigate this risk.

Prioritize and Rank Alerts/Exceptions Streamed to the Dashboard

Because there are tons of raw data, it is important to have a mechanism by which important exceptions/behaviors are proactively pushed to the information consumers. A business rule can be codified, which detects the alert pattern of interest. It can be coded into a program, using database-stored procedures, which can crawl through the fact tables and detect patterns that need immediate attention. This way, information finds the business user as opposed to the business user polling the fact tables for the occurrence of critical patterns.

Enrich the Dashboard with Business-User Comments

When the same dashboard information is presented to multiple business users, a small text box can be provided that can capture the comments from an end-user's perspective. This can often be tagged to the dashboard to put the information in context, adding perspective to the structured KPIs being rendered.

Present Information in Three Different Levels

Information can be presented in three layers depending on the granularity of the information: the visual dashboard level, the static report level, and the self-service cube level. When a user navigates the dashboard, a simple set of 8 to 12 KPIs can be presented, which would give a sense of what is going well and what is not.

Pick the Right Visual Construct Using Dashboard Design Principles

In presenting information in a dashboard, some information is presented best with bar charts, some with time series line graphs, and when presenting correlations, a scatter plot is useful. Sometimes merely rendering it as simple tables is effective. Once the dashboard design principles are explicitly documented, all the developers working on the front end can adhere to the same principles while rendering the reports and dashboard.

Provide for Guided Analytics

In a typical organization, business users can be at various levels of analytical maturity. The capability of the dashboard can be used to guide the “average” business user to access the same navigational path as that of an analytically savvy business user.